

MATH 3150 — Exam 1 Study Guide

Methods, templates, worked examples, and practice problems

(Weeks 1–5: $\mathbb{N}, \mathbb{Q}, \mathbb{R}$, induction, sup/inf, completeness, absolute value, limits of sequences, monotone sequences, subsequences, Cauchy sequences)

Exam 1: Tuesday, October 13, in class (12:30–1:45, MONT 225)

Contents

1	How to use this guide	3
2	Logic and proof techniques	4
2.1	The basic proof moves	4
2.2	Negating quantifiers	5
2.3	Double inclusion (HW1 P1)	5
2.4	Induction (HW1 P2, P3)	5
2.5	Contradiction and irrationality (HW1 P4, Week 2)	8
3	Supremum, infimum, completeness	9
3.1	Definitions (state these exactly)	9
3.2	Archimedean property and density	10
3.3	Computing sup and inf of explicit sets	10
3.4	Standard sup/inf proofs: templates	12
3.5	Monotonicity of sup	14
4	Absolute value	14
5	Limits of sequences: the ε–N definition	15
5.1	Limits equal to $\pm\infty$	18
6	Limit theorems (Week 3–4)	19
6.1	The quotient rule and the “bounded away from zero” lemma	19
6.2	More limit tools	20
6.3	Binomial-theorem estimates (Week 4 style)	20
6.4	Basic limits to memorize	22
7	Monotone sequences and the Monotone Convergence Theorem	22
8	Subsequences and Cauchy sequences (Week 5)	25
8.1	Subsequence facts you need	25
8.2	Definition and intuition	25
8.3	Methods	26

8.4	Worked examples	27
8.5	The long proof: the Cauchy criterion, every step	29
8.6	The Nested Interval Theorem	32
9	Common mistakes that cost points	33
10	Practice problems	35
11	Solutions	37
12	Practice Exam 1 (75 minutes)	41
13	Practice Exam 1: solutions	42

1 How to use this guide

Each section follows the same pattern: **definitions you must be able to state word for word**, then **the method**, then **worked examples**. Section 10 has practice problems, and their solutions are in Section 11. Try each problem on paper before you look at the solution.

Exam 1 scope. The syllabus says Weeks 1–4, as tested in HW1–HW4. **Cauchy sequences will also be tested**, so Section 8 covers them in full, along with the subsequence facts that the Cauchy proofs need.

- HW1: set identities, induction, rational/irrational arguments.
- HW2: density of $\mathbb{Q}$ and of the irrationals, sup/inf, the completeness axiom, $\sup(A + B)$.
- HW3: triangle and reverse triangle inequality, ε – N limits, $x_n \rightarrow \infty$, monotone sequences.
- HW4 and Week 4 lectures: limit laws, basic limits, limits equal to $\pm\infty$, the Monotone Convergence Theorem.
- Quiz 1: $\sup\{xy\} = \sup A \cdot \sup B$ for positive sets (worked in full in Example 3.13).
- Week 5: subsequences, Bolzano–Weierstrass, **Cauchy sequences**, and the **Nested Interval Theorem** (Section 8.6).
- Not tested: the Week 6 topology material.

Exam vs. quiz. The quiz was one proof in 20 minutes. The exam is 75 minutes and will probably have several problems: expect one or two definitions or statements, a sup/inf proof, an ε – N proof, a monotone or recursive sequence, and a Cauchy question. Section 12 is a practice exam in that format.

The single most important skill: match the *shape* of the claim to the proof technique.

Claim looks like...	Reach for...
$S = T$ (sets)	Double inclusion, element chasing
“for all $n \geq n_0$ ”	Induction
“ x is irrational”, “no such ...”	Contradiction
“is it always true?” (suspect false)	One counterexample
$M = \sup S$	(i) M is an upper bound, (ii) $\forall \varepsilon > 0 \exists s \in S, s > M - \varepsilon$
$\sup A \leq \text{something}$	Show that “something” is an upper bound of A
$\lim x_n = L$	ε – N : scratch work, then a forward write-up
$\lim x_n = \infty$	$\forall M \exists N: x_n > M$ for $n \geq N$
x_n converges (limit unknown)	Monotone Convergence Theorem
$ a \leq c$	Show $-c \leq a \leq c$

Definitions cheat sheet (memorize word for word)

Upper bound	M with $s \leq M$ for all $s \in S$
$\sup S$	upper bound M such that no $M' < M$ is an upper bound $\iff$ upper bound and $\forall \varepsilon > 0 \exists s \in S : s > M - \varepsilon$
$\inf S$	lower bound m such that no $m' > m$ is a lower bound $\iff$ lower bound and $\forall \varepsilon > 0 \exists s \in S : s < m + \varepsilon$
Completeness	nonempty + bounded above $\Rightarrow$ sup exists in $\mathbb{R}$
Archimedean	$\forall x \in \mathbb{R} \exists n \in \mathbb{N} : n > x$; equivalently $\forall \varepsilon > 0 \exists n : \frac{1}{n} < \varepsilon$
Density	$x < y \Rightarrow \exists r \in \mathbb{Q}$ and $\exists z \notin \mathbb{Q}$ with $x < r, z < y$
$ x \leq a$	$\iff -a \leq x \leq a$ (for $a \geq 0$)
$x_n \rightarrow L$	$\forall \varepsilon > 0 \exists N \forall n \geq N : x_n - L < \varepsilon$
$x_n \not\rightarrow L$	$\exists \varepsilon > 0 \forall N \exists n \geq N : x_n - L \geq \varepsilon$
$x_n \rightarrow \infty$	$\forall M \exists N \forall n \geq N : x_n > M$
Bounded seq.	$\exists M > 0 \forall n : x_n \leq M$
Increasing	$x_n \leq x_{n+1}$ for all n (decreasing: $x_n \geq x_{n+1}$)
MCT	bounded + monotone $\Rightarrow$ convergent (to sup or inf of the terms)
Cauchy	$\forall \varepsilon > 0 \exists N \forall n, m \geq N : x_n - x_m < \varepsilon$
Not Cauchy	$\exists \varepsilon > 0 \forall N \exists n, m \geq N : x_n - x_m \geq \varepsilon$
Subsequence	$(x_{n_k})_k$ with $n_1 < n_2 < n_3 < \dots$ (so $n_k \geq k$)
Cauchy criterion	in $\mathbb{R}$: (x_n) converges $\iff (x_n)$ is Cauchy
B-W	every bounded sequence has a convergent subsequence
NIT	closed, bounded, nested $[a_n, b_n] \Rightarrow \bigcap_n [a_n, b_n] \neq \emptyset$; a single point if $b_n - a_n \rightarrow 0$

2 Logic and proof techniques

2.1 The basic proof moves

- **Direct:** assume the hypotheses and chain implications to the conclusion.
- **Contrapositive:** $P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$. Example: “ n^2 even $\Rightarrow n$ even” is easiest as “ n odd $\Rightarrow n^2$ odd”.
- **Contradiction:** assume $\neg(\text{claim})$ and derive something false.
- **Proving “ $a \leq b$ ” via ε :** if $a < b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$. (Otherwise take $\varepsilon = a - b > 0$.) Likewise, if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$. **This is how uniqueness of limits and many sup facts are proved.**
- **“For every” statements:** begin with “Let ... be arbitrary.” **“There exists” statements:** construct or exhibit the object.

2.2 Negating quantifiers

To negate a statement, swap $\forall \leftrightarrow \exists$ and negate the inner condition.

$$\neg(\forall \varepsilon > 0 \exists N \forall n \geq N : |x_n - L| < \varepsilon) \iff \exists \varepsilon > 0 \forall N \exists n \geq N : |x_n - L| \geq \varepsilon.$$

So “ $x_n \not\rightarrow L$ ” means: *some* ε is a bad tolerance, and the sequence keeps returning to distance $\geq \varepsilon$ from L no matter how far out you go. You use this to prove divergence (Example 5.8).

2.3 Double inclusion (HW1 P1)

Method 2.1. To prove $S = T$, prove $S \subseteq T$ and $T \subseteq S$. For $S \subseteq T$: “Let $x \in S$ be arbitrary. ... Therefore $x \in T$.” Split into cases wherever the definition involves “or” (a union) or “not (and)” (a complement of an intersection).

Example 2.2 (De Morgan-type identity). $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Quiz-length version. ($\subseteq$) Let $x \in A \setminus (B \cup C)$. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$. Hence $x \in A \setminus B$ and $x \in A \setminus C$.

($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$. Then $x \in A$, $x \notin B$, and $x \notin C$, so $x \notin B \cup C$. Hence $x \in A \setminus (B \cup C)$. $\square$

Full proof, step by step. Step 1 (unpack the definitions you will use).

$$x \in A \setminus D \iff (x \in A \text{ and } x \notin D), \quad x \in D \cup E \iff (x \in D \text{ or } x \in E).$$

Negating the second one gives $x \notin D \cup E \iff (x \notin D \text{ and } x \notin E)$. This negation of an “or” is the whole content of the identity.

Step 2 ($\subseteq$: start with an arbitrary element). Let $x \in A \setminus (B \cup C)$ be arbitrary. By the definition of set difference, $x \in A$ and $x \notin B \cup C$.

Step 3 (negate the union). By Step 1, $x \notin B \cup C$ means $x \notin B$ and $x \notin C$.

Step 4 (repackage). From $x \in A$ and $x \notin B$ we get $x \in A \setminus B$. From $x \in A$ and $x \notin C$ we get $x \in A \setminus C$. So x is in both, i.e. $x \in (A \setminus B) \cap (A \setminus C)$. Since x was arbitrary, $A \setminus (B \cup C) \subseteq (A \setminus B) \cap (A \setminus C)$.

Step 5 ($\supseteq$: arbitrary element of the right side). Let $x \in (A \setminus B) \cap (A \setminus C)$ be arbitrary. By the definition of intersection, $x \in A \setminus B$ and $x \in A \setminus C$, so $x \in A$, $x \notin B$, and $x \notin C$.

Step 6 (rebuild the union). Since $x \notin B$ and $x \notin C$, Step 1 gives $x \notin B \cup C$. Together with $x \in A$, this says $x \in A \setminus (B \cup C)$. So $(A \setminus B) \cap (A \setminus C) \subseteq A \setminus (B \cup C)$.

Step 7 (conclude). Both inclusions hold, so the sets are equal. $\square$

Why no cases here? There is no “or” among the *assumptions*. The only “or” is inside $x \notin B \cup C$, and negating it turns it into an “and”. In Practice Problem 4 the “or” is an assumption, so you must split into cases.

2.4 Induction (HW1 P2, P3)

Method 2.3. 1. **Base case:** check $P(n_0)$ directly.

2. **Write out** $P(n+1)$ by replacing every n with $n+1$. This is your target, so write it down first.

3. **Inductive step:** assume $P(n)$. Find the piece of $P(n+1)$ that is exactly $P(n)$ and substitute the hypothesis.

4. Finish with algebra.

If the *starting* index of a sum also moves with n , add and subtract the boundary terms that don't match (HW1 P3 trick).

Example 2.4 (Bernoulli's inequality, used in Week 4). For $x \geq -1$ and $n \in \mathbb{N}$: $(1+x)^n \geq 1+nx$.

Proof. Base case $n = 1$: $1+x \geq 1+x$. ✓

Inductive step: assume $(1+x)^n \geq 1+nx$. Since $1+x \geq 0$, we may multiply both sides by $1+x$ without flipping the inequality:

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

because $nx^2 \geq 0$. □

Full proof, step by step. Fix $x \geq -1$. Let $P(n)$ be the statement $(1+x)^n \geq 1+nx$. We prove $P(n)$ for all $n \in \mathbb{N}$ by induction.

Step 1 (base case $n = 1$). LHS = $(1+x)^1 = 1+x$ and RHS = $1+1 \cdot x = 1+x$. Since $1+x \geq 1+x$, $P(1)$ holds. ✓

Step 2 (write the target $P(n+1)$).

$$(1+x)^{n+1} \geq 1 + (n+1)x.$$

Step 3 (peel off one factor). Assume $P(n)$. Then $(1+x)^{n+1} = (1+x)^n \cdot (1+x)$.

Step 4 (use the hypothesis $x \geq -1$). Since $x \geq -1$, we have $1+x \geq 0$. Multiplying both sides of $P(n)$ by the *nonnegative* number $1+x$ keeps the direction of the inequality:

$$(1+x)^n(1+x) \geq (1+nx)(1+x).$$

Step 5 (expand).

$$(1+nx)(1+x) = 1+x+nx+nx^2 = 1+(n+1)x+nx^2.$$

Step 6 (drop the extra term). Since $n \geq 1$ and $x^2 \geq 0$, we have $nx^2 \geq 0$, so $1+(n+1)x+nx^2 \geq 1+(n+1)x$.

Step 7 (chain and conclude). Combining Steps 3–6,

$$(1+x)^{n+1} = (1+x)^n(1+x) \geq (1+nx)(1+x) = 1+(n+1)x+nx^2 \geq 1+(n+1)x,$$

which is $P(n+1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$. □

Note: the hypothesis $x \geq -1$ is used exactly once, to multiply an inequality by $1+x \geq 0$. Look for where each hypothesis is used.

What goes wrong for $x < -1$? Take $x = -4$, $n = 3$: $(1-4)^3 = -27$ but $1+3(-4) = -11$, and $-27 < -11$. So the claim really can fail when $x < -1$, and Step 4 is exactly where the proof breaks.

Example 2.5 (Product identity, HW1 P2 style). For $n \geq 2$: $\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$.

Key step: $\frac{n+1}{2n} \cdot \frac{(n+1)^2-1}{(n+1)^2} = \frac{n+1}{2n} \cdot \frac{n(n+2)}{(n+1)^2} = \frac{n+2}{2(n+1)}$, using $(n+1)^2 - 1 = n(n+2)$ (difference of squares).

Full proof, step by step. Let $P(n)$ be the statement $\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$. We prove $P(n)$ for all $n \geq 2$ by induction.

Step 1 (base case $n = 2$). The product has a single factor, $k = 2$:

$$\text{LHS} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4}, \quad \text{RHS} = \frac{2+1}{2 \cdot 2} = \frac{3}{4}. \checkmark$$

Step 2 (write the target $P(n+1)$). Replace every n by $n+1$:

$$\prod_{k=2}^{n+1} \left(1 - \frac{1}{k^2}\right) = \frac{(n+1)+1}{2(n+1)} = \frac{n+2}{2(n+1)}.$$

This is what we must reach.

Step 3 (peel off the last factor). Assume $P(n)$. The product up to $n+1$ is the product up to n , times the $k = n+1$ factor:

$$\prod_{k=2}^{n+1} \left(1 - \frac{1}{k^2}\right) = \underbrace{\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right)}_{\text{this is } P(n)} \cdot \left(1 - \frac{1}{(n+1)^2}\right).$$

Step 4 (substitute the inductive hypothesis).

$$= \frac{n+1}{2n} \cdot \left(1 - \frac{1}{(n+1)^2}\right).$$

Step 5 (common denominator in the last factor).

$$1 - \frac{1}{(n+1)^2} = \frac{(n+1)^2}{(n+1)^2} - \frac{1}{(n+1)^2} = \frac{(n+1)^2 - 1}{(n+1)^2}.$$

Step 6 (factor the numerator). Use $a^2 - b^2 = (a-b)(a+b)$ with $a = n+1$, $b = 1$:

$$(n+1)^2 - 1 = ((n+1) - 1)((n+1) + 1) = n(n+2).$$

(Check by expanding: $(n+1)^2 - 1 = n^2 + 2n + 1 - 1 = n^2 + 2n = n(n+2)$.)

Step 7 (multiply and cancel).

$$\frac{n+1}{2n} \cdot \frac{n(n+2)}{(n+1)^2} = \frac{(n+1)n(n+2)}{2n(n+1)^2} = \frac{\cancel{n}(n+1)(n+2)}{2\cancel{n}(n+1)^2} = \frac{n+2}{2(n+1)}.$$

Cancelling is allowed because $n \neq 0$ and $n+1 \neq 0$ (as $n \geq 2$).

Step 8 (conclude). This is exactly the target from Step 2, so $P(n) \Rightarrow P(n+1)$. Together with the base case, $P(n)$ holds for all $n \geq 2$. $\square$

Sanity check $n = 3$: $\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{9}\right) = \frac{3}{4} \cdot \frac{8}{9} = \frac{24}{36} = \frac{2}{3}$, and $\frac{3+1}{2 \cdot 3} = \frac{4}{6} = \frac{2}{3}$. $\checkmark$

Where does $\frac{n+1}{2n}$ come from? (telescoping) Each factor splits as $1 - \frac{1}{k^2} = \frac{k^2-1}{k^2} = \frac{k-1}{k} \cdot \frac{k+1}{k}$, so

$$\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \left(\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{n-1}{n}\right) \left(\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n}\right) = \frac{1}{n} \cdot \frac{n+1}{2} = \frac{n+1}{2n}.$$

Almost everything cancels in each bracket. This is a good way to *guess* the formula. The induction proof above is what makes it rigorous.

2.5 Contradiction and irrationality (HW1 P4, Week 2)

Method 2.6. Assume the opposite. Solve for the piece you know is irrational. Closure of $\mathbb{Q}$ under $+$, $-$, $\times$, $\div$ then forces that piece to be rational, which is a contradiction.

Example 2.7 ($\sqrt{2} \notin \mathbb{Q}$, from lecture: know this proof cold). Suppose $\sqrt{2} = p/q$ in lowest terms. Then $p^2 = 2q^2$, so p^2 is even. If $p = 2k + 1$ were odd, then $p^2 = 2(2k^2 + 2k) + 1$ would be odd. So p is even, $p = 2k$. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ and q is even too. This contradicts lowest terms.

Lemma used twice. For an integer m : if m^2 is even, then m is even.

Proof (contrapositive). Suppose m is odd, so $m = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$m^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is $2 \cdot (\text{integer}) + 1$, so m^2 is odd. Hence “ m odd $\Rightarrow m^2$ odd”, which is equivalent to “ m^2 even $\Rightarrow m$ even”. $\square$

Full proof, step by step. Step 1 (assume the opposite). Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$.

Step 2 (write it in lowest terms). Then $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{Z}$ and $q \neq 0$. By cancelling common factors we may assume the fraction is in lowest terms, i.e. p and q have *no common factor greater than 1*. (Every fraction can be reduced, for example $\frac{6}{4} = \frac{3}{2}$.) This extra assumption is what we will contradict.

Step 3 (clear the square root). Square both sides and multiply by q^2 :

$$2 = \frac{p^2}{q^2} \implies p^2 = 2q^2.$$

Step 4 (p is even). $p^2 = 2q^2$ is $2 \cdot (\text{integer})$, so p^2 is even. By the Lemma, p is even.

Step 5 (substitute $p = 2k$). Since p is even, $p = 2k$ for some $k \in \mathbb{Z}$. Plug into $p^2 = 2q^2$:

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies q^2 = 2k^2.$$

Step 6 (q is even). $q^2 = 2k^2$ is $2 \cdot (\text{integer})$, so q^2 is even. By the Lemma again, q is even.

Step 7 (contradiction). Now 2 divides both p and q , so they share the common factor 2. This contradicts Step 2 (lowest terms).

Step 8 (conclude). The assumption in Step 1 is false, so $\sqrt{2} \notin \mathbb{Q}$. $\square$

Why lowest terms? Without it there is no contradiction. Finding that p and q are both even would just mean the fraction could be reduced. Lowest terms is the “trap” that the evenness argument springs.

Common point-losers:

- Saying “ p^2 even, so p even” with no justification. Prove the Lemma, or at least cite it.
- Forgetting to say $q \neq 0$ and to assume lowest terms at the start.
- Concluding “ p even” from “ $p^2 = 2q^2$ ” without first noting that p^2 is even.

Sanity check: why doesn’t this “prove” $\sqrt{4} \notin \mathbb{Q}$? Run the same steps: $p^2 = 4q^2$, so p is even, $p = 2k$. Then $4k^2 = 4q^2$, so $q^2 = k^2$. This says *nothing* about q being even, so no contradiction arises, which is correct because $\sqrt{4} = 2 \in \mathbb{Q}$. The factor 2 appearing exactly once in $p^2 = 2q^2$ is what makes Step 6 work.

Same template for $\sqrt{3}$: replace “even/odd” with “divisible by 3 or not.” If $3 \nmid m$, then $m = 3k + 1$ or $m = 3k + 2$, and in both cases m^2 leaves remainder 1 when divided by 3. So $3 \mid m^2 \Rightarrow 3 \mid m$, and the rest of the proof goes through unchanged.

Example 2.8 (rational $\pm$ irrational). Let $r \in \mathbb{Q}$ and $y \notin \mathbb{Q}$. Then

- (a) $r + y \notin \mathbb{Q}$ and $r - y \notin \mathbb{Q}$;
- (b) if also $r \neq 0$, then $ry \notin \mathbb{Q}$ and $y/r \notin \mathbb{Q}$.

Tool: closure of $\mathbb{Q}$. If $a = \frac{p}{q}$ and $b = \frac{s}{t}$ with $p, q, s, t \in \mathbb{Z}$ and $q, t \neq 0$, then

$$a + b = \frac{pt + sq}{qt}, \quad a - b = \frac{pt - sq}{qt}, \quad ab = \frac{ps}{qt}, \quad \frac{a}{b} = \frac{pt}{qs} \quad (\text{if } b \neq 0, \text{ i.e. } s \neq 0).$$

Each is an integer over a nonzero integer, so it is rational. This is the only fact about $\mathbb{Q}$ the proof uses.

Proof of (a), the sum. Step 1 (assume the opposite). Suppose, for contradiction, that $r + y \in \mathbb{Q}$. Give it a name: $z := r + y$, so $z \in \mathbb{Q}$.

Step 2 (solve for the irrational piece). Subtract r from both sides: $y = z - r$.

Step 3 (apply closure). Both z and r are rational, so $z - r$ is rational by closure under subtraction. Hence $y \in \mathbb{Q}$.

Step 4 (contradiction). This contradicts the hypothesis $y \notin \mathbb{Q}$. So the assumption in Step 1 was false, and $r + y \notin \mathbb{Q}$. $\square$

Proof of (a), the difference. Suppose $r - y = z \in \mathbb{Q}$. Solving for y gives $y = r - z$, which is rational by closure. This contradicts $y \notin \mathbb{Q}$, so $r - y \notin \mathbb{Q}$. $\square$

Proof of (b), the product. Step 1. Suppose, for contradiction, that $ry \in \mathbb{Q}$. Call it $w := ry$, so $w \in \mathbb{Q}$.

Step 2. Solve for y . Because $r \neq 0$ we may divide by r : $y = \frac{w}{r}$.

Step 3. w and r are rational and $r \neq 0$, so w/r is rational by closure under division. Hence $y \in \mathbb{Q}$.

Step 4. This contradicts $y \notin \mathbb{Q}$, so $ry \notin \mathbb{Q}$.

For the quotient, note $1/r \in \mathbb{Q}$ and $1/r \neq 0$, and $y/r = (1/r) \cdot y$. The product case with $1/r$ in place of r gives $y/r \notin \mathbb{Q}$. $\square$

Where is $r \neq 0$ used? Only in Step 2 of (b), to divide by r . It is really needed: if $r = 0$, then $ry = 0 \in \mathbb{Q}$, so the claim is false.

Concrete check. $3 + \sqrt{2} \notin \mathbb{Q}$: if $3 + \sqrt{2} = z \in \mathbb{Q}$, then $\sqrt{2} = z - 3 \in \mathbb{Q}$, which contradicts the previous example. Likewise $5\sqrt{2} \notin \mathbb{Q}$, because otherwise $\sqrt{2} = \frac{5\sqrt{2}}{5} \in \mathbb{Q}$.

Counterexample question: is irrational + irrational always irrational? No: $\sqrt{2} + (-\sqrt{2}) = 0$. Another one is $\sqrt{2} + (1 - \sqrt{2}) = 1$, where $1 - \sqrt{2} \notin \mathbb{Q}$ by part (a). Is irrational $\times$ irrational always irrational? No: $\sqrt{2} \cdot \sqrt{2} = 2$ and $\sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$.

Why the proof above fails here: Step 3 needs r to be rational to conclude $z - r \in \mathbb{Q}$. If r is irrational, closure says nothing, and the argument breaks down.

3 Supremum, infimum, completeness

3.1 Definitions (state these exactly)

Definition 3.1. Let $S \subseteq \mathbb{R}$ be nonempty.

- M is an **upper bound** of S if $s \leq M$ for all $s \in S$.
- $M = \sup S$ if M is an upper bound of S and no $M' < M$ is an upper bound of S . (It is the *least* upper bound.)

- $m = \inf S$ if m is a lower bound of S and no $m' > m$ is a lower bound of S .
- $\max S$ is an upper bound that *belongs to* S . If $\max S$ exists, then $\sup S = \max S$.

Theorem 3.2 (Completeness Axiom). Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. (It follows that every nonempty set bounded below has an infimum.)

Theorem 3.3 (ε -characterization of sup, the workhorse). $M = \sup S$ if and only if

- (i) $s \leq M$ for all $s \in S$, and
- (ii) for every $\varepsilon > 0$ there is $s \in S$ with $s > M - \varepsilon$.

The same holds for inf with the inequalities flipped: (ii) becomes $\forall \varepsilon > 0 \exists s \in S$ with $s < m + \varepsilon$.

Two facts used constantly:

- If t is *any* upper bound of S , then $\sup S \leq t$. (This is the “least” part.)
- $\sup S - \varepsilon$ is *not* an upper bound, so some element of S exceeds it.

3.2 Archimedean property and density

Theorem 3.4 (Archimedean property). For any $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $n > x$. Equivalently: for any $\varepsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \varepsilon$.

Proof idea (contradiction plus completeness): if $\mathbb{N}$ were bounded above, let $M = \sup \mathbb{N}$. Then $M - 1$ is not an upper bound, so some $n > M - 1$. But then $n + 1 > M$ with $n + 1 \in \mathbb{N}$, contradiction.

Theorem 3.5 (Density). If $x < y$, there is $r \in \mathbb{Q}$ with $x < r < y$, and there is an irrational z with $x < z < y$.

Proof idea for $\mathbb{Q}$: pick n with $1/n < y - x$. Let $m = \lfloor nx \rfloor + 1$. Then $nx < m \leq nx + 1 < ny$, so $x < m/n < y$.

Proof idea for irrationals (HW2 P1): either take rationals $x < r_1 < r_2 < y$ and $z = r_1 + \frac{r_2 - r_1}{\sqrt{2}}$, or apply density of $\mathbb{Q}$ to $x/\sqrt{2} < r < y/\sqrt{2}$ with $r \neq 0$ and take $z = r\sqrt{2}$.

3.3 Computing sup and inf of explicit sets

Method 3.6. 1. List the first few elements to guess the sup and inf.

- 2. If the candidate is *attained* (it is an element of S), show it is a bound. It is then the max or min, and you are done.
- 3. If it is *not attained*, show it is a bound, then use Theorem 3.3(ii) together with the Archimedean property.

Example 3.7. $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\} = \{0, \frac{1}{2}, \frac{2}{3}, \dots\}$. Then $\inf S = \min S = 0$ (at $n = 1$) and $\sup S = 1$ (not attained).

Proof that $\sup S = 1$. (i) $1 - \frac{1}{n} < 1$ for all n , so 1 is an upper bound. (ii) Let $\varepsilon > 0$. By the Archimedean property, choose n with $\frac{1}{n} < \varepsilon$. Then $1 - \frac{1}{n} > 1 - \varepsilon$, and $1 - \frac{1}{n} \in S$. By Theorem 3.3, $\sup S = 1$. $\square$

Full proof, step by step. Step 1 ($\inf S = \min S = 0$). For every $n \in \mathbb{N}$, $n \geq 1$, so $\frac{1}{n} \leq 1$ and $1 - \frac{1}{n} \geq 0$. Thus 0 is a lower bound of S . Also $0 = 1 - \frac{1}{1} \in S$. A lower bound that belongs to the set is the minimum, so $\inf S = \min S = 0$.

Step 2 (1 is an upper bound). For every $n \in \mathbb{N}$, $\frac{1}{n} > 0$, so $1 - \frac{1}{n} < 1$. Hence $s \leq 1$ for all $s \in S$. This is part (i) of Theorem 3.3.

Step 3 (start part (ii)). Let $\varepsilon > 0$ be arbitrary. We must find an element of S that is larger than $1 - \varepsilon$.

Step 4 (scratch: what n works?). $1 - \frac{1}{n} > 1 - \varepsilon \iff \frac{1}{n} < \varepsilon$. So we need n with $\frac{1}{n} < \varepsilon$.

Step 5 (Archimedean property). Such an $n \in \mathbb{N}$ exists by the Archimedean property. Fix one.

Step 6 (verify). Then $s := 1 - \frac{1}{n} \in S$ and $s = 1 - \frac{1}{n} > 1 - \varepsilon$.

Step 7 (conclude). Steps 2–6 verify (i) and (ii) of Theorem 3.3, so $\sup S = 1$.

Step 8 (not attained). If $1 \in S$ then $1 = 1 - \frac{1}{n}$ for some n , so $\frac{1}{n} = 0$, which is impossible. So $1 \notin S$ and S has no maximum. $\square$

Example 3.8 (HW2 P2). $S_1 = \{\frac{1}{n}\}$: $\sup = \max = 1$, and $\inf = 0$ is not attained. $S_2 = \{\frac{(-1)^n}{n}\}$: split into odd and even n . Then $\sup S_2 = \max = \frac{1}{2}$ (at $n = 2$) and $\inf S_2 = \min = -1$ (at $n = 1$).

Full proof for S_1 , step by step. *Step 1* ($\sup S_1 = \max S_1 = 1$). For every $n \in \mathbb{N}$, $n \geq 1$, so $\frac{1}{n} \leq 1$. So 1 is an upper bound, and $1 = \frac{1}{1} \in S_1$. An upper bound that belongs to the set is the maximum, so $\sup S_1 = \max S_1 = 1$.

Step 2 (0 is a lower bound). For every n , $\frac{1}{n} > 0$. So $0 \leq s$ for all $s \in S_1$.

Step 3 (the ε part of the inf). Let $\varepsilon > 0$. By the Archimedean property there is $n \in \mathbb{N}$ with $\frac{1}{n} < \varepsilon = 0 + \varepsilon$, and $\frac{1}{n} \in S_1$. By Theorem 3.3 (inf version), $\inf S_1 = 0$.

Step 4 (not attained). $\frac{1}{n} = 0$ has no solution, so $0 \notin S_1$ and S_1 has no minimum. $\square$

Full proof for S_2 , step by step. *Step 1 (list and split).* The elements are $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \dots$

- If n is even, then $\frac{(-1)^n}{n} = \frac{1}{n}$, and $n \geq 2$, so $0 < \frac{1}{n} \leq \frac{1}{2}$.

- If n is odd, then $\frac{(-1)^n}{n} = -\frac{1}{n}$, and $n \geq 1$, so $-1 \leq -\frac{1}{n} < 0$.

Step 2 (upper bound $\frac{1}{2}$). Even terms are $\leq \frac{1}{2}$ by Step 1. Odd terms are $< 0 < \frac{1}{2}$. So every element is $\leq \frac{1}{2}$.

Step 3 (attained). $n = 2$ gives $\frac{(-1)^2}{2} = \frac{1}{2} \in S_2$. So $\sup S_2 = \max S_2 = \frac{1}{2}$.

Step 4 (lower bound -1). Odd terms are ≥ -1 by Step 1. Even terms are $> 0 > -1$. So every element is ≥ -1 .

Step 5 (attained). $n = 1$ gives $-1 \in S_2$. So $\inf S_2 = \min S_2 = -1$. $\square$

Lesson: when the candidate is attained, you never need ε . “Bound + element of the set” finishes it.

Example 3.9. $S = (a, b)$: $\sup S = b$ and $\inf S = a$, and neither is attained. For (ii): given $\varepsilon > 0$ (with $\varepsilon < b - a$ if needed), the point $x = b - \varepsilon/2 \in S$ satisfies $x > b - \varepsilon$.

Full proof that $\sup(a, b) = b$ (for $a < b$), step by step. *Step 1 (upper bound).* If $x \in (a, b)$ then $a < x < b$, so $x \leq b$. Thus b is an upper bound.

Step 2 (start part (ii)). Let $\varepsilon > 0$ be arbitrary. We need $x \in (a, b)$ with $x > b - \varepsilon$.

Step 3 (handle large ε). The point $b - \frac{\varepsilon}{2}$ might fall at or below a when ε is large. To avoid this, let $\delta = \min\{\varepsilon, b - a\} > 0$, and set $x = b - \frac{\delta}{2}$.

Step 4 ($x \in (a, b)$). Since $\delta > 0$, $x < b$. Since $\delta \leq b - a$, we have $\frac{\delta}{2} < b - a$, so $x = b - \frac{\delta}{2} > b - (b - a) = a$. So $a < x < b$.

Step 5 ($x > b - \varepsilon$). Since $\delta \leq \varepsilon$, we have $\frac{\delta}{2} < \varepsilon$, so $x = b - \frac{\delta}{2} > b - \varepsilon$.

Step 6 (conclude). By Theorem 3.3, $\sup(a, b) = b$.

Step 7 (not attained). $b \notin (a, b)$ since $b < b$ is false. So (a, b) has no maximum. $\square$

The proof that $\inf(a, b) = a$ is the mirror image: use $x = a + \frac{\delta}{2}$ with the same δ .

3.4 Standard sup/inf proofs: templates

Example 3.10 ($\sup S \leq \inf T$, HW2 P3). If $s \leq t$ for all $s \in S, t \in T$: fix t . Then t is an upper bound of S , so $\sup S \leq t$. Since t was arbitrary, $\sup S$ is a lower bound of T , so $\sup S \leq \inf T$.

Full proof, step by step. Let $S, T \subseteq \mathbb{R}$ be nonempty with $s \leq t$ for all $s \in S, t \in T$.

Step 1 (sup S exists). $T \neq \emptyset$, so pick some $t_0 \in T$. By hypothesis $s \leq t_0$ for all $s \in S$, so S is bounded above. Since $S \neq \emptyset$, the Completeness Axiom gives that $\sup S$ exists. Similarly, any $s_0 \in S$ is a lower bound of T , so $\inf T$ exists.

Step 2 (fix an arbitrary t). Let $t \in T$ be arbitrary.

Step 3 (t is an upper bound of S). By hypothesis $s \leq t$ for every $s \in S$. That is the definition of “ t is an upper bound of S ”.

Step 4 (“least” upper bound). $\sup S$ is less than or equal to every upper bound of S . So $\sup S \leq t$.

Step 5 (turn it around). Since $t \in T$ was arbitrary, Step 4 says $\sup S \leq t$ for all $t \in T$. So $\sup S$ is a lower bound of T .

Step 6 (“greatest” lower bound). $\inf T$ is greater than or equal to every lower bound of T . So $\sup S \leq \inf T$. $\square$

Pattern: “fix one, bound, then let it vary” is used twice. You apply the “least” property to S and then the “greatest” property to T .

Example 3.11 ($\sup(A + B) = \sup A + \sup B$, HW2 P4). ($\leq$) Since $a + b \leq \sup A + \sup B$, the right side is an upper bound. ($\geq$) Given $\varepsilon > 0$, pick $a > \sup A - \frac{\varepsilon}{2}$ and $b > \sup B - \frac{\varepsilon}{2}$. Then $a + b > \sup A + \sup B - \varepsilon$. **Splitting ε into halves** is the standard trick.

Full proof, step by step. Let A, B be nonempty and bounded above, and $A + B = \{a + b : a \in A, b \in B\}$. Write $\alpha = \sup A$ and $\beta = \sup B$ (these exist by completeness). We show $\sup(A + B) = \alpha + \beta$ using Theorem 3.3.

Step 1 ($A + B$ is nonempty). Pick $a_0 \in A$ and $b_0 \in B$. Then $a_0 + b_0 \in A + B$.

Step 2 (upper bound). Let $x \in A + B$ be arbitrary. Then $x = a + b$ for some $a \in A, b \in B$. Since $a \leq \alpha$ and $b \leq \beta$, adding gives $x = a + b \leq \alpha + \beta$. So $\alpha + \beta$ is an upper bound of $A + B$. In particular $\sup(A + B)$ exists.

Step 3 (start part (ii)). Let $\varepsilon > 0$. We need some $x \in A + B$ with $x > \alpha + \beta - \varepsilon$.

Step 4 (split ε). $\frac{\varepsilon}{2} > 0$, so $\alpha - \frac{\varepsilon}{2}$ is not an upper bound of A . Hence there is $a \in A$ with $a > \alpha - \frac{\varepsilon}{2}$. In the same way there is $b \in B$ with $b > \beta - \frac{\varepsilon}{2}$.

Step 5 (add). Then $a + b \in A + B$ and

$$a + b > \left(\alpha - \frac{\varepsilon}{2}\right) + \left(\beta - \frac{\varepsilon}{2}\right) = \alpha + \beta - \varepsilon.$$

Step 6 (conclude). Steps 2 and 5 are (i) and (ii) of Theorem 3.3, so $\sup(A + B) = \alpha + \beta$. $\square$

Why $\frac{\varepsilon}{2}$ and not ε ? With $a > \alpha - \varepsilon$ and $b > \beta - \varepsilon$ you only get $a + b > \alpha + \beta - 2\varepsilon$, which is too weak. Each of the two pieces gets half of the budget.

Example 3.12 ($\sup(cA) = c\sup A$ for $c > 0$). Let $\alpha = \sup A$. For $a \in A, ca \leq c\alpha$, so $c\alpha$ is an upper bound of cA . Given $\varepsilon > 0$, pick $a \in A$ with $a > \alpha - \varepsilon/c$. Then $ca > c\alpha - \varepsilon$. (What changes if $c < 0$? Then $\sup(cA) = c\inf A$.)

Full proof, step by step. Let A be nonempty and bounded above, $c > 0$, and $cA = \{ca : a \in A\}$. Let $\alpha = \sup A$.

Step 1 (upper bound). Let $x \in cA$, so $x = ca$ for some $a \in A$. Since $a \leq \alpha$ and $c > 0$, multiplying by c keeps the inequality: $ca \leq c\alpha$. So $c\alpha$ is an upper bound of cA .

Step 2 (start part (ii)). Let $\varepsilon > 0$. We want $x \in cA$ with $x > c\alpha - \varepsilon$.

Step 3 (scratch: rescale ε). If $x = ca$, then $ca > c\alpha - \varepsilon \iff a > \alpha - \frac{\varepsilon}{c}$ (divide by $c > 0$). So we need an element of A within $\frac{\varepsilon}{c}$ of α .

Step 4 (use $\sup A$). $\frac{\varepsilon}{c} > 0$, so $\alpha - \frac{\varepsilon}{c}$ is not an upper bound of A . So there is $a \in A$ with $a > \alpha - \frac{\varepsilon}{c}$.

Step 5 (multiply back). Multiply by $c > 0$: $ca > c\alpha - \varepsilon$, and $ca \in cA$.

Step 6 (conclude). By Theorem 3.3, $\sup(cA) = c\alpha = c \sup A$. $\square$

Where is $c > 0$ used? In Steps 1, 3 and 5, every time we multiply or divide an inequality by c . If $c < 0$ each of those inequalities flips, upper bounds turn into lower bounds, and you get $\sup(cA) = c \inf A$ instead.

Example 3.13 (Quiz 1: $\sup(AB) = \sup A \cdot \sup B$ for positive sets). Let $A, B \subseteq (0, \infty)$ be nonempty and bounded, with $\alpha = \sup A$, $\beta = \sup B$, and $C = \{xy : x \in A, y \in B\}$. Prove $\sup C = \alpha\beta$.

Idea. ($\leq$) is the usual “upper bound” step. For ($\geq$), the instructor’s solution avoids ε : it **divides by one variable to turn the problem into a sup of A** , then does the same for B . This is “fix one, bound, let it vary” from the $\sup S \leq \inf T$ template, done twice.

Full proof, step by step. *Step 1 ($\alpha\beta$ is an upper bound of C).* Let $z \in C$, so $z = xy$ with $x \in A$, $y \in B$. Then $0 < x \leq \alpha$ and $0 < y \leq \beta$. Multiplying inequalities between *positive* numbers, $xy \leq \alpha y \leq \alpha\beta$. So $\alpha\beta$ is an upper bound of C , $\sup C$ exists by completeness ($C \neq \emptyset$), and $\sup C \leq \alpha\beta$.

Step 2 (fix y , bound A). Fix $y \in B$. For every $x \in A$, $xy \in C$, so $xy \leq \sup C$. Since $y > 0$, $x \leq \frac{\sup C}{y}$. So $\frac{\sup C}{y}$ is an upper bound of A , and since α is the *least* upper bound, $\alpha \leq \frac{\sup C}{y}$.

Step 3 (let y vary, bound B). Step 2 gives $y \leq \frac{\sup C}{\alpha}$ for every $y \in B$ (here $\alpha > 0$ because A contains a positive number). So $\frac{\sup C}{\alpha}$ is an upper bound of B , and therefore $\beta \leq \frac{\sup C}{\alpha}$, i.e. $\alpha\beta \leq \sup C$.

Step 4. Steps 1 and 3 give $\sup C = \alpha\beta$. $\square$

Where is positivity used? In every multiplication or division (Steps 1, 2, 3). Without it the result fails: for $A = B = \{-1, -2\}$ we get $\sup A = \sup B = -1$, but $C = \{1, 2, 4\}$ and $\sup C = 4 \neq 1$.

ε version (harder). You can also pick $x > \alpha - \delta$, $y > \beta - \delta$ and expand $(\alpha - \delta)(\beta - \delta) > \alpha\beta - \delta(\alpha + \beta)$, but you then have to choose δ small enough to keep both factors positive. The division trick above is cleaner. **Expect a variation on the exam**, e.g. $\sup\{1/x : x \in A\} = 1/\inf A$ (Practice exam, Problem 3).

Example 3.14 ($\inf S = -\sup(-S)$). $m \leq s$ for all $s \in S \iff -s \leq -m$ for all $s \in S$. So lower bounds of S are exactly the negatives of upper bounds of $-S$, and the greatest lower bound corresponds to the least upper bound.

Full proof, step by step. Let S be nonempty and bounded below, and let $-S = \{-s : s \in S\}$.

Step 1 (key equivalence). For any real m :

$$\begin{aligned} m \text{ is a lower bound of } S &\iff m \leq s \text{ for all } s \in S \\ &\iff -s \leq -m \text{ for all } s \in S \\ &\iff -m \text{ is an upper bound of } -S. \end{aligned}$$

(Multiplying by -1 flips inequalities.)

Step 2 (sup($-S$) exists). S has some lower bound m , so by Step 1 $-m$ is an upper bound of $-S$. Also $-S \neq \emptyset$. By completeness, $\beta := \sup(-S)$ exists.

Step 3 ($-\beta$ is a lower bound of S). β is an upper bound of $-S$, i.e. $-(\beta)$ is an upper bound of $-S$. By Step 1 (with $m = -\beta$), $-\beta$ is a lower bound of S .

Step 4 ($-\beta$ is the greatest). Let m be any lower bound of S . By Step 1, $-m$ is an upper bound of $-S$. Since β is the least upper bound, $\beta \leq -m$, so $m \leq -\beta$.

Step 5 (conclude). $-\beta$ is a lower bound of S and every lower bound is $\leq -\beta$. By definition, $\inf S = -\beta = -\sup(-S)$. $\square$

Use: this lets you get every inf fact from the matching sup fact, and also shows that completeness for sups gives completeness for infs.

3.5 Monotonicity of sup

If $A \subseteq B$, then $\sup A \leq \sup B$ and $\inf A \geq \inf B$. (Any upper bound of B is an upper bound of A .)

4 Absolute value

Theorem 4.1 (Properties).

1. $|xy| = |x||y|$ and $|-x| = |x|$.
2. For $a \geq 0$: $|x| \leq a \iff -a \leq x \leq a$. (**Most useful fact.**)
3. Triangle inequality: $|x + y| \leq |x| + |y|$.
4. Reverse triangle inequality: $||x| - |y|| \leq |x - y|$.
5. $|x - a| < \delta \iff a - \delta < x < a + \delta$.

Method 4.2 (Proving the triangle inequality). Add $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$ to get $-(|x| + |y|) \leq x + y \leq |x| + |y|$, then apply property 2.

Method 4.3 (Reverse triangle inequality, HW3 P1). Write $x = (x - y) + y$, so $|x| \leq |x - y| + |y|$, i.e. $|x| - |y| \leq |x - y|$. Swap x and y to get $-(|x| - |y|) \leq |x - y|$. Then property 2 gives the result.

Method 4.4 (The add-and-subtract trick). To compare a and b through c , write $a - b = (a - c) + (c - b)$ and apply the triangle inequality. This is how uniqueness of limits and the product rule are proved.

Example 4.5 (Solving equations with absolute values). Solve $|x - 3| + |x - 1| = 3$. Break the line at the points 1 and 3:

- $x < 1$: $(3 - x) + (1 - x) = 3 \Rightarrow x = \frac{1}{2}$. $\checkmark$
- $1 \leq x < 3$: $(3 - x) + (x - 1) = 2 \neq 3$. No solution.
- $x \geq 3$: $(x - 3) + (x - 1) = 3 \Rightarrow x = \frac{7}{2}$. $\checkmark$

Full solution, step by step. Step 1 (find the break points). $|x - 3|$ changes formula at $x = 3$, and $|x - 1|$ changes formula at $x = 1$. These points split $\mathbb{R}$ into $(-\infty, 1)$, $[1, 3)$ and $[3, \infty)$.

Step 2 (sign table). Use $|u| = u$ if $u \geq 0$ and $|u| = -u$ if $u < 0$:

	$x - 3$	$x - 1$
$x < 1$	< 0 , so $ x - 3 = 3 - x$	< 0 , so $ x - 1 = 1 - x$
$1 \leq x < 3$	< 0 , so $ x - 3 = 3 - x$	≥ 0 , so $ x - 1 = x - 1$
$x \geq 3$	≥ 0 , so $ x - 3 = x - 3$	> 0 , so $ x - 1 = x - 1$

Step 3 (case $x < 1$). The equation becomes $(3 - x) + (1 - x) = 3$, i.e. $4 - 2x = 3$, so $x = \frac{1}{2}$. Check that it lies in the case: $\frac{1}{2} < 1$. ✓

Step 4 (case $1 \leq x < 3$). The equation becomes $(3 - x) + (x - 1) = 3$, i.e. $2 = 3$. This is false, so there is no solution in this interval.

Step 5 (case $x \geq 3$). The equation becomes $(x - 3) + (x - 1) = 3$, i.e. $2x - 4 = 3$, so $x = \frac{7}{2}$. Check: $\frac{7}{2} \geq 3$. ✓

Step 6 (verify in the original equation). $|\frac{1}{2} - 3| + |\frac{1}{2} - 1| = \frac{5}{2} + \frac{1}{2} = 3$ and $|\frac{7}{2} - 3| + |\frac{7}{2} - 1| = \frac{1}{2} + \frac{5}{2} = 3$.

Step 7 (answer). The solution set is $\{\frac{1}{2}, \frac{7}{2}\}$. □

Always check that a case's answer lies in that case's interval. If it doesn't, throw it away.
Geometric check: $|x - 3| + |x - 1|$ is the total distance from x to 1 and to 3. Between 1 and 3 this total is always 2. You must move $\frac{1}{2}$ outside the interval on either side to get 3.

5 Limits of sequences: the ε - N definition

Definition 5.1. $x_n \rightarrow L$ means: for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n - L| < \varepsilon$ for all $n \geq N$.

Method 5.2 (The ε - N recipe). 1. **Guess** L (leading terms, or intuition from calculus).

2. **Scratch work:** simplify $|x_n - L|$ to a single fraction.

3. **Overestimate** it by something simpler, such as $\frac{C}{n}$. You make a fraction larger by making the denominator smaller or the numerator larger. You only need an upper bound.

4. Solve $\frac{C}{n} < \varepsilon$, which gives $n > \frac{C}{\varepsilon}$, so $N > \frac{C}{\varepsilon}$ (Archimedean).

5. **Write it forwards:** "Let $\varepsilon > 0$. Choose $N > \dots$. Then for $n \geq N$: $|x_n - L| = \dots \leq \dots < \varepsilon$."

Example 5.3 (HW3 P2). $\frac{n+2}{3n+5} \rightarrow \frac{1}{3}$.

Scratch work (not part of the proof): $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{3(3n+5)} = \frac{1}{9n+15} < \frac{1}{9n}$. We want $\frac{1}{9n} < \varepsilon$, i.e. $n > \frac{1}{9\varepsilon}$.

Proof. Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{1}{9\varepsilon}$. For $n \geq N$, $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15} < \frac{1}{9n} \leq \frac{1}{9N} < \varepsilon$. □

Full proof, step by step. Step 1 (simplify $|x_n - L|$, scratch). Use the common denominator $3(3n+5)$:

$$\frac{n+2}{3n+5} - \frac{1}{3} = \frac{3(n+2) - (3n+5)}{3(3n+5)} = \frac{3n+6-3n-5}{9n+15} = \frac{1}{9n+15}.$$

This is positive, so $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15}$.

Step 2 (overestimate, scratch). Since $9n+15 > 9n > 0$, $\frac{1}{9n+15} < \frac{1}{9n}$. (A smaller denominator gives a larger fraction.)

Step 3 (solve for n , scratch). $\frac{1}{9n} < \varepsilon \iff 9n\varepsilon > 1 \iff n > \frac{1}{9\varepsilon}$.

Step 4 (the proof starts here: let $\varepsilon > 0$). Let $\varepsilon > 0$ be arbitrary.

Step 5 (choose N). By the Archimedean property, choose $N \in \mathbb{N}$ with $N > \frac{1}{9\varepsilon}$. Note that N depends only on ε .

Step 6 (verify for every $n \geq N$). Let $n \geq N$. Then $9n \geq 9N > \frac{1}{\varepsilon}$, so $\frac{1}{9n} \leq \frac{1}{9N} < \varepsilon$. Therefore

$$\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15} < \frac{1}{9n} \leq \frac{1}{9N} < \varepsilon.$$

Step 7 (conclude). Since $\varepsilon > 0$ was arbitrary, $\frac{n+2}{3n+5} \rightarrow \frac{1}{3}$ by definition. $\square$

Example 5.4 (Lecture). $\frac{2n-1}{n+1} \rightarrow 2$: $\left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1} < \frac{3}{n}$. Choose $N > \frac{3}{\varepsilon}$.

Full proof, step by step. Step 1 (simplify, scratch).

$$\frac{2n-1}{n+1} - 2 = \frac{2n-1-2(n+1)}{n+1} = \frac{-3}{n+1}, \quad \text{so} \quad \left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1}.$$

Step 2 (overestimate, scratch). $n+1 > n$, so $\frac{3}{n+1} < \frac{3}{n}$. Then $\frac{3}{n} < \varepsilon \iff n > \frac{3}{\varepsilon}$.

Step 3 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$. By the Archimedean property choose $N \in \mathbb{N}$ with $N > \frac{3}{\varepsilon}$.

Step 4 (verify). For $n \geq N$:

$$\left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1} < \frac{3}{n} \leq \frac{3}{N} < \varepsilon.$$

Step 5 (conclude). So $\frac{2n-1}{n+1} \rightarrow 2$. $\square$

Watch the sign: the difference is $\frac{-3}{n+1}$, which is negative. The absolute value removes the minus sign, so don't skip it.

Example 5.5 (A numerator that needs bounding). $\frac{3n^2+n}{n^2+1} \rightarrow 3$: $\left| \frac{3n^2+n}{n^2+1} - 3 \right| = \frac{|n-3|}{n^2+1} \leq \frac{n+3}{n^2} \leq \frac{4n}{n^2} = \frac{4}{n}$ for $n \geq 1$ (since $3 \leq 3n$). Choose $N > \frac{4}{\varepsilon}$. **Bound the numerator above and the denominator below.**

Full proof, step by step. Step 1 (simplify, scratch).

$$\frac{3n^2+n}{n^2+1} - 3 = \frac{3n^2+n-3n^2-3}{n^2+1} = \frac{n-3}{n^2+1}, \quad \text{so} \quad \left| \frac{3n^2+n}{n^2+1} - 3 \right| = \frac{|n-3|}{n^2+1}.$$

Step 2 (numerator: bound above). By the triangle inequality, $|n-3| \leq |n| + |-3| = n+3$. Since $n \geq 1$, $3 \leq 3n$, so $n+3 \leq 4n$.

Step 3 (denominator: bound below). $n^2+1 > n^2$, so $\frac{1}{n^2+1} < \frac{1}{n^2}$.

Step 4 (combine).

$$\frac{|n-3|}{n^2+1} < \frac{4n}{n^2} = \frac{4}{n}.$$

(For $n=3$ the left side is 0, which is still $< \frac{4}{n}$.)

Step 5 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{4}{\varepsilon}$.

Step 6 (verify). For $n \geq N$: $\left| \frac{3n^2+n}{n^2+1} - 3 \right| < \frac{4}{n} \leq \frac{4}{N} < \varepsilon$.

Step 7 (conclude). So $\frac{3n^2+n}{n^2+1} \rightarrow 3$. $\square$

Why not solve $\frac{|n-3|}{n^2+1} < \varepsilon$ exactly? You could, but it needs the quadratic formula and cases. Overestimating by $\frac{4}{n}$ is valid and much faster. You only need *some* N that works, not the smallest one.

Example 5.6 (Conjugate trick, lecture). $\frac{\sqrt{n^2+n}}{n} \rightarrow 1$: $\frac{\sqrt{n^2+n}-n}{n} = \frac{n}{n(\sqrt{n^2+n}+n)} = \frac{1}{\sqrt{n^2+n}+n} < \frac{1}{n}$. Choose $N > \frac{1}{\varepsilon}$. **Use** $\sqrt{a}-\sqrt{b} = \frac{a-b}{\sqrt{a}+\sqrt{b}}$.

Full proof, step by step. Step 1 (common denominator, scratch). $\frac{\sqrt{n^2+n}}{n} - 1 = \frac{\sqrt{n^2+n}-n}{n}$.
Step 2 (multiply by the conjugate).

$$\sqrt{n^2+n}-n = \frac{(\sqrt{n^2+n}-n)(\sqrt{n^2+n}+n)}{\sqrt{n^2+n}+n} = \frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n} = \frac{n}{\sqrt{n^2+n}+n}.$$

Step 3 (substitute and cancel).

$$\frac{\sqrt{n^2+n}-n}{n} = \frac{n}{n(\sqrt{n^2+n}+n)} = \frac{1}{\sqrt{n^2+n}+n}.$$

This is positive, so it equals its absolute value.

Step 4 (overestimate). $\sqrt{n^2+n} > 0$, so $\sqrt{n^2+n}+n > n$ and $\frac{1}{\sqrt{n^2+n}+n} < \frac{1}{n}$.

Step 5 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$.

Step 6 (verify). For $n \geq N$:

$$\left| \frac{\sqrt{n^2+n}}{n} - 1 \right| = \frac{1}{\sqrt{n^2+n}+n} < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Step 7 (conclude). So $\frac{\sqrt{n^2+n}}{n} \rightarrow 1$. □

Why the conjugate? $\sqrt{n^2+n}-n$ is a difference of two large, nearly equal numbers, so you can't see its size. After multiplying by the conjugate, the n^2 terms cancel.

Example 5.7 (Bounded over growing). $\frac{\sin n}{n} \rightarrow 0$: $\left| \frac{\sin n}{n} \right| \leq \frac{1}{n} < \varepsilon$ when $n > \frac{1}{\varepsilon}$. (You cannot use the quotient rule, because $\lim \sin n$ does not exist.)

Full proof, step by step. Step 1 (the only fact about $\sin$). $|\sin t| \leq 1$ for all real t . In particular $|\sin n| \leq 1$.

Step 2 (bound the term). For $n \in \mathbb{N}$: $\left| \frac{\sin n}{n} - 0 \right| = \frac{|\sin n|}{n} \leq \frac{1}{n}$.

Step 3 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$.

Step 4 (verify). For $n \geq N$: $\left| \frac{\sin n}{n} - 0 \right| \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon$.

Step 5 (conclude). So $\frac{\sin n}{n} \rightarrow 0$. □

Example 5.8 (Proving divergence). $x_n = (-1)^n$ has no limit. Suppose $x_n \rightarrow L$. Take $\varepsilon = \frac{1}{2}$ and get N . Choose an even $n \geq N$ and an odd $m \geq N$. Then $|1-L| < \frac{1}{2}$ and $|-1-L| < \frac{1}{2}$, so $2 = |1-(-1)| \leq |1-L| + |L-(-1)| < 1$, which is a contradiction.

Full proof, step by step. Step 1 (assume the opposite). Suppose, for contradiction, that $(-1)^n \rightarrow L$ for some $L \in \mathbb{R}$.

Step 2 (choose one specific ε). The definition holds for *every* $\varepsilon > 0$, so it holds for $\varepsilon = \frac{1}{2}$. So there is $N \in \mathbb{N}$ with $|(-1)^n - L| < \frac{1}{2}$ for all $n \geq N$.

Step 3 (pick one even and one odd index past N). Take $n = 2N$ (even) and $m = 2N + 1$ (odd). Both are $\geq N$.

Step 4 (read off the two inequalities). $(-1)^{2N} = 1$ and $(-1)^{2N+1} = -1$, so

$$|1 - L| < \frac{1}{2} \quad \text{and} \quad |-1 - L| < \frac{1}{2}.$$

Step 5 (add and subtract L , then triangle inequality).

$$2 = |1 - (-1)| = |(1 - L) + (L - (-1))| \leq |1 - L| + |L + 1| < \frac{1}{2} + \frac{1}{2} = 1.$$

(Here $|L + 1| = |-1 - L|$.)

Step 6 (contradiction). Step 5 says $2 < 1$, which is false. So no such L exists, and $(-1)^n$ diverges. $\square$

Why $\varepsilon = \frac{1}{2}$? The terms 1 and -1 are 2 apart. Any $\varepsilon \leq 1$ works, because two points within ε of L are less than $2\varepsilon \leq 2$ apart. $\varepsilon = \frac{1}{2}$ leaves some room.

5.1 Limits equal to $\pm\infty$

Definition 5.9. $x_n \rightarrow \infty$ means: for every M there is N such that $x_n > M$ for all $n \geq N$. Also, $x_n \rightarrow -\infty$ means $-x_n \rightarrow \infty$.

Method 5.10. Find a *simpler lower bound* $x_n \geq g(n)$ with $g(n) \rightarrow \infty$ obviously, such as $g(n) = cn$. Then choose N so that $g(n) > M$.

Example 5.11 (Lecture). $\frac{n^4 - 4n}{n^2} \rightarrow \infty$: for $n \geq 2$, $n^4 - 4n \geq n^3$ (since $n^3(n - 1) \geq n^3 \geq 4n$ when $n \geq 2$), so $\frac{n^4 - 4n}{n^2} \geq n$. Given M , choose $N > \max(M, 2)$.

Full proof, step by step. Step 1 (simplify). $\frac{n^4 - 4n}{n^2} = n^2 - \frac{4}{n}$. We'll work with the original form, since the lower bound is easy to see there.

Step 2 (lower bound on the numerator, for $n \geq 2$). We claim $n^4 - 4n \geq n^3$, i.e. $n^4 - n^3 \geq 4n$. Factor: $n^4 - n^3 = n^3(n - 1)$. For $n \geq 2$ we have $n - 1 \geq 1$, so $n^3(n - 1) \geq n^3 = n \cdot n^2 \geq n \cdot 4 = 4n$ (since $n^2 \geq 4$).

Step 3 (simpler lower bound for x_n). Dividing Step 2 by $n^2 > 0$: for $n \geq 2$, $\frac{n^4 - 4n}{n^2} \geq \frac{n^3}{n^2} = n$.

Step 4 (let M be arbitrary and choose N). Let $M \in \mathbb{R}$. By the Archimedean property choose $N \in \mathbb{N}$ with $N > \max(M, 2)$.

Step 5 (verify). Let $n \geq N$. Then $n \geq 2$, so Step 3 applies, and $\frac{n^4 - 4n}{n^2} \geq n \geq N > M$.

Step 6 (conclude). For every M there is such an N , so $\frac{n^4 - 4n}{n^2} \rightarrow \infty$. $\square$

Why the max? $N > M$ gives the size condition, and $N \geq 2$ makes Step 3 valid. Taking the max gets both at once.

Example 5.12 (HW3 P3: $\infty + \text{bounded} = \infty$). Say $|y_n| \leq B$. Then $x_n + y_n \geq x_n - B$. Given K , apply $x_n \rightarrow \infty$ with the **threshold** $K + B$ to get N . For $n \geq N$, $x_n + y_n > K + B - B = K$.

Full proof, step by step. Assume $x_n \rightarrow \infty$ and (y_n) is bounded. We show $x_n + y_n \rightarrow \infty$.

Step 1 (name the bound). Since (y_n) is bounded, there is $B > 0$ with $|y_n| \leq B$ for all n . In particular $-B \leq y_n$ for all n .

Step 2 (lower bound for the sum). Adding x_n gives $x_n + y_n \geq x_n - B$ for all n .

Step 3 (let K be arbitrary). Let $K \in \mathbb{R}$. We must find N with $x_n + y_n > K$ for all $n \geq N$.

Step 4 (scratch: what do we need from x_n ?). By Step 2, it is enough that $x_n - B > K$, i.e. $x_n > K + B$.

Step 5 (apply $x_n \rightarrow \infty$ with threshold $K + B$). The definition of $x_n \rightarrow \infty$ holds for every real threshold, in particular for $M = K + B$. So there is N with $x_n > K + B$ for all $n \geq N$.

Step 6 (verify). For $n \geq N$: $x_n + y_n \geq x_n - B > (K + B) - B = K$.

Step 7 (conclude). K was arbitrary, so $x_n + y_n \rightarrow \infty$. $\square$

Only the lower bound $y_n \geq -B$ is used. So the result holds whenever (y_n) is merely bounded below.

6 Limit theorems (Week 3–4)

Theorem 6.1. Suppose $x_n \rightarrow x$ and $y_n \rightarrow y$. Then:

1. **Uniqueness:** limits are unique.
2. **Bounded:** a convergent sequence is bounded.
3. $ax_n + by_n \rightarrow ax + by$.
4. $x_n y_n \rightarrow xy$.
5. $x_n / y_n \rightarrow x / y$ if $y \neq 0$.
6. **Squeeze:** if $a_n \leq z_n \leq b_n$ and $a_n, b_n \rightarrow L$, then $z_n \rightarrow L$.
7. **Order:** if $x_n \leq y_n$ for all n (large), then $x \leq y$. In particular, $x_n \geq 0 \Rightarrow x \geq 0$. **Strict inequality is not preserved:** $\frac{1}{n} > 0$ but the limit is 0.

Proofs to know (short ones are exam favorites):

- *Uniqueness:* $|x - x'| \leq |x - x_n| + |x_n - x'| < 2\varepsilon$ for every ε , so $x = x'$.
- *Bounded:* with $\varepsilon = 1$, $|x_n| < |x| + 1$ for $n \geq N$. Take $M = \max\{|x_1|, \dots, |x_{N-1}|, |x| + 1\}$.
- *Sum:* $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$, using $N = \max(N_1, N_2)$.
- *Product:* $|x_n y_n - xy| = |(x_n - x)y_n + x(y_n - y)| \leq M|x_n - x| + |x||y_n - y|$, where $|y_n| \leq M$ because convergent sequences are bounded.
- *Squeeze:* $L - \varepsilon < a_n \leq z_n \leq b_n < L + \varepsilon$.
- *Order, by contradiction:* if $x_n \geq 0$ and $x < 0$, take $\varepsilon = -x/2$. Then $x_n < x + \varepsilon = x/2 < 0$ for large n , contradiction.

6.1 The quotient rule and the “bounded away from zero” lemma

Theorem 6.2 (Lemma). If $y_n \rightarrow y \neq 0$, then there is N_0 such that $|y_n| > \frac{|y|}{2}$ for all $n \geq N_0$.

Proof. Apply the definition with $\varepsilon = \frac{|y|}{2}$. Then $|y_n| \geq |y| - |y - y_n| > |y| - \frac{|y|}{2} = \frac{|y|}{2}$ by the reverse triangle inequality. $\square$

Quotient rule sketch: by the product rule, it suffices to show $\frac{1}{y_n} \rightarrow \frac{1}{y}$. For $n \geq N_0$,

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \frac{|y - y_n|}{|y_n||y|} < \frac{2}{|y|^2} |y_n - y|,$$

which can be made $< \varepsilon$ by choosing N with $|y_n - y| < \frac{|y|^2 \varepsilon}{2}$.

6.2 More limit tools

- **Zero times bounded:** if $x_n \rightarrow 0$ and $|y_n| \leq M$, then $x_n y_n \rightarrow 0$, because $|x_n y_n| \leq M|x_n|$. (This does *not* need y_n to converge.)
- **Shifts don't matter:** $x_n \rightarrow L \iff x_{n+1} \rightarrow L \iff x_{n+k} \rightarrow L$. Changing finitely many terms doesn't change the limit.
- **Reciprocals and infinity:** if $x_n \rightarrow \infty$, then $\frac{1}{x_n} \rightarrow 0$. If $x_n > 0$ and $x_n \rightarrow 0$, then $\frac{1}{x_n} \rightarrow \infty$.
- **Infinity arithmetic:** $x_n \rightarrow \infty$ and $y_n \geq c > 0$ eventually $\Rightarrow x_n y_n \rightarrow \infty$. $x_n \rightarrow \infty$ and y_n bounded below $\Rightarrow x_n + y_n \rightarrow \infty$. $\infty - \infty$, $0 \cdot \infty$ and $\frac{\infty}{\infty}$ are **indeterminate**: for example $n - n = 0$, $(n+1) - n = 1$, and $n^2 - n \rightarrow \infty$.
- **Comparison for infinity:** if $x_n \leq y_n$ and $x_n \rightarrow \infty$, then $y_n \rightarrow \infty$.
- **The “max of the N ’s”:** when you need two conditions at once, take $N = \max(N_1, N_2)$.
- **ε -splitting:** to get a total of $< \varepsilon$ from two pieces, make each piece $< \frac{\varepsilon}{2}$. With a constant C in front, make the piece $< \frac{\varepsilon}{C}$.

6.3 Binomial-theorem estimates (Week 4 style)

For $x \geq 0$ and $n \geq 2$: $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k \geq 1 + nx$ and also $(1+x)^n \geq \binom{n}{2} x^2 = \frac{n(n-1)}{2} x^2$. Keep only the one term you need.

Example 6.3 ($\frac{n}{2^n} \rightarrow 0$). $2^n = (1+1)^n \geq \binom{n}{2} = \frac{n(n-1)}{2}$, so for $n \geq 2$, $0 < \frac{n}{2^n} \leq \frac{2n}{n(n-1)} = \frac{2}{n-1} \rightarrow 0$. Squeeze.

Full proof, step by step. Step 1 (binomial theorem). For $n \geq 2$,

$$2^n = (1+1)^n = \sum_{k=0}^n \binom{n}{k} \geq \binom{n}{2} = \frac{n(n-1)}{2},$$

because every term of the sum is ≥ 0 and we kept only the $k=2$ term.

Step 2 (bound the sequence). For $n \geq 2$, $n-1 > 0$, so taking reciprocals (both sides positive) gives $\frac{1}{2^n} \leq \frac{2}{n(n-1)}$. Multiplying by $n > 0$:

$$0 < \frac{n}{2^n} \leq \frac{2n}{n(n-1)} = \frac{2}{n-1}.$$

Step 3 ($\frac{2}{n-1} \rightarrow 0$). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > 1 + \frac{2}{\varepsilon}$ (also $N \geq 2$). For $n \geq N$, $n-1 \geq N-1 > \frac{2}{\varepsilon}$, so $0 < \frac{2}{n-1} < \varepsilon$.

Step 4 (squeeze). $0 \leq \frac{n}{2^n} \leq \frac{2}{n-1}$ for $n \geq 2$, and both outer sequences tend to 0. By the Squeeze Theorem, $\frac{n}{2^n} \rightarrow 0$. (Or directly: for $n \geq N$, $|\frac{n}{2^n}| \leq \frac{2}{n-1} < \varepsilon$.) $\square$

Why $\binom{n}{2}$ and not $1+n$? Bernoulli gives $2^n \geq 1+n$, so $\frac{n}{2^n} \leq \frac{n}{n+1}$, which does *not* go to 0. You need a term of degree 2 in n to beat the n on top.

Example 6.4 ($a^{1/n} \rightarrow 1$ for $a > 1$). Let $d_n = a^{1/n} - 1 > 0$. Then $a = (1+d_n)^n \geq 1 + nd_n$, so $0 < d_n \leq \frac{a-1}{n} \rightarrow 0$. For $0 < a < 1$, apply this to $\frac{1}{a}$ and take reciprocals.

Full proof, step by step. Step 1 (define the error term). Let $a > 1$ and $d_n := a^{1/n} - 1$. Since $a > 1$, $a^{1/n} > 1$ (if $a^{1/n} \leq 1$ then $a = (a^{1/n})^n \leq 1$). So $d_n > 0$.

Step 2 (rewrite a). $a^{1/n} = 1 + d_n$, so $a = (1 + d_n)^n$.

Step 3 (Bernoulli). Since $d_n > 0 \geq -1$, Bernoulli's inequality gives $(1 + d_n)^n \geq 1 + nd_n$. So $a \geq 1 + nd_n$.

Step 4 (solve for d_n). $nd_n \leq a - 1$, so $0 < d_n \leq \frac{a-1}{n}$.

Step 5 (ε - N). Let $\varepsilon > 0$ and choose $N > \frac{a-1}{\varepsilon}$. For $n \geq N$: $|a^{1/n} - 1| = d_n \leq \frac{a-1}{n} \leq \frac{a-1}{N} < \varepsilon$. So $a^{1/n} \rightarrow 1$.

Step 6 (the case $a = 1$). $1^{1/n} = 1$ for all n , so the limit is 1.

Step 7 (the case $0 < a < 1$). Then $b := \frac{1}{a} > 1$, so $b^{1/n} \rightarrow 1$ by Step 5. Since $a^{1/n} = \frac{1}{b^{1/n}}$ and the limit $1 \neq 0$, the quotient rule gives $a^{1/n} \rightarrow \frac{1}{1} = 1$. $\square$

Example 6.5 ($n^{1/n} \rightarrow 1$). Let $d_n = n^{1/n} - 1 \geq 0$. Then $n = (1 + d_n)^n \geq \frac{n(n-1)}{2}d_n^2$, so $d_n^2 \leq \frac{2}{n-1}$ and $0 \leq d_n \leq \sqrt{\frac{2}{n-1}} \rightarrow 0$.

Full proof, step by step. Step 1 (error term). Let $d_n := n^{1/n} - 1$. Since $n \geq 1$, $n^{1/n} \geq 1$ (as in Step 1 above), so $d_n \geq 0$.

Step 2 (binomial theorem, keep the $k = 2$ term). For $n \geq 2$,

$$n = (1 + d_n)^n = \sum_{k=0}^n \binom{n}{k} d_n^k \geq \binom{n}{2} d_n^2 = \frac{n(n-1)}{2} d_n^2,$$

since every term is ≥ 0 (here $d_n \geq 0$ is used).

Step 3 (solve for d_n). Divide by $\frac{n(n-1)}{2} > 0$: $d_n^2 \leq \frac{2n}{n(n-1)} = \frac{2}{n-1}$. Since $d_n \geq 0$, taking square roots gives $0 \leq d_n \leq \sqrt{\frac{2}{n-1}}$.

Step 4 (scratch). $\sqrt{\frac{2}{n-1}} < \varepsilon \iff \frac{2}{n-1} < \varepsilon^2 \iff n > 1 + \frac{2}{\varepsilon^2}$.

Step 5 (ε - N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > 1 + \frac{2}{\varepsilon^2}$ (so also $N \geq 2$). For $n \geq N$:

$$|n^{1/n} - 1| = d_n \leq \sqrt{\frac{2}{n-1}} \leq \sqrt{\frac{2}{N-1}} < \varepsilon.$$

Step 6 (conclude). So $n^{1/n} \rightarrow 1$. $\square$

Why not Bernoulli here? It gives $n \geq 1 + nd_n$, i.e. $d_n \leq \frac{n-1}{n}$, which does not go to 0. The $\binom{n}{2}$ term is needed because the base n grows.

Example 6.6 ($\frac{a^n}{n!} \rightarrow 0$ for any a). Fix an integer $K \geq 2|a|$. For $n > K$, $\frac{|a|^n}{n!} = \frac{|a|^K}{K!} \cdot \prod_{j=K+1}^n \frac{|a|}{j} \leq \frac{|a|^K}{K!} \left(\frac{1}{2}\right)^{n-K} \rightarrow 0$.

Full proof, step by step. Step 1 (reduce to $|a|$). $\left|\frac{a^n}{n!} - 0\right| = \frac{|a|^n}{n!}$, so it is enough to show $\frac{|a|^n}{n!} \rightarrow 0$. If $a = 0$ every term is 0, so assume $a \neq 0$.

Step 2 (fix a cutoff K). By the Archimedean property choose an integer $K \geq 1$ with $K \geq 2|a|$. Then for every $j \geq K + 1$, $j > 2|a|$, so $\frac{|a|}{j} < \frac{1}{2}$.

Step 3 (split the product). For $n > K$, write $|a|^n = |a|^K \cdot |a|^{n-K}$ and $n! = K! \cdot (K+1)(K+2) \cdots n$. So

$$\frac{|a|^n}{n!} = \frac{|a|^K}{K!} \cdot \frac{|a|}{K+1} \cdot \frac{|a|}{K+2} \cdots \frac{|a|}{n}.$$

There are exactly $n - K$ factors after the first.

Step 4 (bound each tail factor). By Step 2 each of those $n - K$ factors is $< \frac{1}{2}$. Let $C := \frac{|a|^K}{K!}$, a constant (it does not depend on n). Then

$$0 < \frac{|a|^n}{n!} \leq C \left(\frac{1}{2}\right)^{n-K} = C 2^K \left(\frac{1}{2}\right)^n.$$

Step 5 (the bound tends to 0). $\left(\frac{1}{2}\right)^n \rightarrow 0$ (basic limit, $|\frac{1}{2}| < 1$), so $C 2^K \left(\frac{1}{2}\right)^n \rightarrow 0$ by the constant multiple rule.

Step 6 (squeeze). $0 < \frac{|a|^n}{n!} \leq C 2^K \left(\frac{1}{2}\right)^n$ for $n > K$. By the Squeeze Theorem (only large n matter), $\frac{|a|^n}{n!} \rightarrow 0$, and therefore $\frac{a^n}{n!} \rightarrow 0$. $\square$

Idea: the first K factors can be large, but there are only finitely many of them, so they form a fixed constant C . After that, each new factor at least halves the term.

6.4 Basic limits to memorize

1. $\frac{1}{n^p} \rightarrow 0$ for $p > 0$.
2. **Ratio of polynomials of equal degree:** the limit is the ratio of leading coefficients. (Divide the top and bottom by n^k .) If the top has lower degree, the limit is 0. If the top has higher degree, the limit is $\pm\infty$.
3. $a^n \rightarrow 0$ if $|a| < 1$. (Proof: write $\frac{1}{|a|} = 1 + x$ and use Bernoulli: $|a|^n \leq \frac{1}{1+nx} \rightarrow 0$.)
4. $1 + a + \dots + a^n = \frac{1-a^{n+1}}{1-a} \rightarrow \frac{1}{1-a}$ if $|a| < 1$.
5. $\frac{\text{bounded}}{\rightarrow \infty} \rightarrow 0$, e.g. $\frac{\sin n}{n}$ and $\frac{(-1)^n}{n}$.
6. If $x_n \rightarrow L$ then $|x_n| \rightarrow |L|$ (reverse triangle inequality).

7 Monotone sequences and the Monotone Convergence Theorem

Definition 7.1. (x_n) is **increasing** if $x_n \leq x_{n+1}$ for all n , **decreasing** if $x_n \geq x_{n+1}$ for all n , and **monotone** if it is one of the two.

Method 7.2 (Showing monotonicity: three tools). 1. Show $x_{n+1} - x_n \geq 0$ (HW3 P4: common denominator, then check the sign).

2. If $x_n > 0$, show $\frac{x_{n+1}}{x_n} \geq 1$.

3. For recursive sequences, use induction: $x_n \leq x_{n+1} \Rightarrow x_{n+1} \leq x_{n+2}$.

Theorem 7.3 (Monotone Convergence Theorem). A bounded monotone sequence converges. If it is increasing and bounded above, then $x_n \rightarrow \sup\{x_n\}$.

Proof. Let $L = \sup\{x_n\}$, which exists by completeness. Let $\varepsilon > 0$. Since $L - \varepsilon$ is not an upper bound, there is N with $x_N > L - \varepsilon$. For $n \geq N$: $L - \varepsilon < x_N \leq x_n \leq L < L + \varepsilon$, so $|x_n - L| < \varepsilon$. $\square$

This proof is short and is exactly Theorem 3.3(ii) plus monotonicity. Be able to reproduce it.

Method 7.4 (Recursive sequences: the 4-step routine). 1. Compute x_1, x_2, x_3 to guess the direction and the limit.

2. Prove a bound by induction.

3. Prove monotonicity (often by induction, or using the bound).

4. By MCT, $x_n \rightarrow L$. Pass to the limit in the recursion using the limit laws, and *also* $x_{n+1} \rightarrow L$ (it is the same sequence shifted). Solve for L and use the bounds to discard wrong roots.

Example 7.5. $x_1 = 1$, $x_{n+1} = \sqrt{2 + x_n}$. Then $x_n \rightarrow 2$.

Proof. Bounded: $0 < x_n < 2$ by induction. Indeed $x_1 = 1$, and if $0 < x_n < 2$ then $0 < x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2$.

Increasing: $x_{n+1} > x_n \iff 2 + x_n > x_n^2 \iff (2 - x_n)(1 + x_n) > 0$, which holds because $0 < x_n < 2$.

Limit: by MCT, $x_n \rightarrow L$ with $1 \leq L \leq 2$. Square the recursion: $x_{n+1}^2 = 2 + x_n$. Take limits using the product and sum laws: $L^2 = 2 + L$, so $(L - 2)(L + 1) = 0$. Since $L \geq 1$, $L = 2$. $\square$

Full proof, step by step. Step 1 (compute and guess). $x_1 = 1$, $x_2 = \sqrt{3} \approx 1.73$, $x_3 = \sqrt{2 + \sqrt{3}} \approx 1.93$. The terms increase, apparently toward 2. (If $L = \sqrt{2 + L}$ then $L = 2$.)

Step 2 (bound by induction: $0 < x_n < 2$ for all n). Base: $0 < x_1 = 1 < 2$. Inductive step: assume $0 < x_n < 2$. Then $2 < 2 + x_n < 4$, and since $\sqrt{\cdot}$ is increasing, $0 < \sqrt{2} < \sqrt{2 + x_n} < \sqrt{4} = 2$, i.e. $0 < x_{n+1} < 2$.

Step 3 (increasing). Fix n . Both x_n and x_{n+1} are positive, so $x_{n+1} > x_n \iff x_{n+1}^2 > x_n^2$. Since $x_{n+1}^2 = 2 + x_n$:

$$x_{n+1} > x_n \iff 2 + x_n - x_n^2 > 0 \iff (2 - x_n)(1 + x_n) > 0.$$

By Step 2, $2 - x_n > 0$ and $1 + x_n > 0$, so the product is positive and $x_{n+1} > x_n$.

Step 4 (MCT). (x_n) is increasing and bounded above by 2, so by the Monotone Convergence Theorem it converges, say $x_n \rightarrow L$.

Step 5 (bounds on L). $1 = x_1 \leq x_n < 2$ for all n , so by the order property of limits $1 \leq L \leq 2$.

Step 6 (pass to the limit). The recursion gives $x_{n+1}^2 = 2 + x_n$ for all n . The shifted sequence (x_{n+1}) also converges to L , so by the product rule $x_{n+1}^2 \rightarrow L^2$. By the sum rule $2 + x_n \rightarrow 2 + L$. By uniqueness of limits, $L^2 = 2 + L$.

Step 7 (solve and discard). $L^2 - L - 2 = (L - 2)(L + 1) = 0$, so $L = 2$ or $L = -1$. Step 5 rules out -1 . Hence $x_n \rightarrow 2$. $\square$

Order matters: Step 6 is only valid *after* Step 4. Without knowing the limit exists, “ $L = \sqrt{2 + L}$ ” is meaningless.

Example 7.6 (A decreasing recursive sequence). $x_1 = 3$, $x_{n+1} = \sqrt{x_n + 2}$. *Bound:* $x_n > 2$ by induction ($x_{n+1} > \sqrt{4} = 2$). *Decreasing:* $x_{n+1} < x_n \iff x_n + 2 < x_n^2 \iff (x_n - 2)(x_n + 1) > 0$, which holds. By MCT, $x_n \rightarrow L \geq 2$ with $L^2 = L + 2$, so $L = 2$.

Full proof, step by step. Step 1 (compute and guess). $x_1 = 3$, $x_2 = \sqrt{5} \approx 2.24$, $x_3 = \sqrt{2 + \sqrt{5}} \approx 2.06$. The terms decrease, apparently toward 2.

Step 2 (bound by induction: $x_n > 2$ for all n). Base: $x_1 = 3 > 2$. Inductive step: assume $x_n > 2$. Then $x_n + 2 > 4$, so $x_{n+1} = \sqrt{x_n + 2} > \sqrt{4} = 2$.

Step 3 (decreasing). Both $x_n, x_{n+1} > 0$, so $x_{n+1} < x_n \iff x_{n+1}^2 < x_n^2 \iff x_n + 2 < x_n^2$, i.e.

$$x_n^2 - x_n - 2 = (x_n - 2)(x_n + 1) > 0.$$

By Step 2, $x_n - 2 > 0$ and $x_n + 1 > 0$, so this holds. Thus $x_{n+1} < x_n$.

Step 4 (MCT). (x_n) is decreasing and bounded below by 2. By the Monotone Convergence Theorem (decreasing version: it converges to $\inf\{x_n\}$), $x_n \rightarrow L$ for some L .

Step 5 (bounds on L). $2 < x_n \leq 3$ for all n , so by the order property $2 \leq L \leq 3$. (Strict $>$ becomes $\geq$ in the limit.)

Step 6 (pass to the limit). $x_{n+1}^2 = x_n + 2$. The left side tends to L^2 (shift and product rule). The right side tends to $L + 2$. So $L^2 = L + 2$.

Step 7 (solve and discard). $(L - 2)(L + 1) = 0$, so $L \in \{2, -1\}$, and Step 5 forces $L = 2$. $\square$

Compare with the previous example. Same recursion, but a different starting point. Starting above the fixed point 2 gives a decreasing sequence, and starting below it gives an increasing one. The bound you prove by induction always points toward the fixed point.

Example 7.7 (e exists). $a_n = \left(1 + \frac{1}{n}\right)^n$ is increasing and bounded above by 3 (binomial expansion compared with $\sum \frac{1}{k!} \leq 1 + \sum \frac{1}{2^{k-1}}$). So it converges, and its limit is called e . **MCT proves existence without knowing the value.**

Full proof, step by step. Step 1 (expand with the binomial theorem).

$$a_n = \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k}.$$

For $k \geq 1$,

$$\binom{n}{k} \frac{1}{n^k} = \frac{n(n-1) \cdots (n-k+1)}{k! n^k} = \frac{1}{k!} \cdot \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-k+1}{n} = \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

So

$$a_n = \sum_{k=0}^n \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right),$$

where the empty product (for $k = 0$) is 1.

Step 2 (compare a_n with a_{n+1} term by term). Fix $k \leq n$. For each $j \geq 0$, $\frac{j}{n+1} \leq \frac{j}{n}$, so $1 - \frac{j}{n+1} \geq 1 - \frac{j}{n} \geq 0$ (as $j \leq k-1 < n$). All factors are nonnegative, so the products compare the same way:

$$\frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n+1}\right) \geq \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

So term k of a_{n+1} is $\geq$ term k of a_n , for each $k = 0, \dots, n$.

Step 3 (increasing). a_{n+1} also has one extra term ($k = n+1$), which is ≥ 0 . Adding up Step 2 over $k = 0, \dots, n$ gives $a_{n+1} \geq a_n$.

Step 4 (drop the products). Each factor $1 - \frac{j}{n} \in [0, 1]$, so each product is ≤ 1 and $a_n \leq \sum_{k=0}^n \frac{1}{k!}$.

Step 5 ($k! \geq 2^{k-1}$ for $k \geq 1$). By induction: $1! = 1 = 2^0$, and if $k! \geq 2^{k-1}$ then $(k+1)! = (k+1)k! \geq 2 \cdot 2^{k-1} = 2^k$ (since $k+1 \geq 2$).

Step 6 (geometric bound).

$$a_n \leq 1 + \sum_{k=1}^n \frac{1}{k!} \leq 1 + \sum_{k=1}^n \frac{1}{2^{k-1}} = 1 + \frac{1 - (1/2)^n}{1 - 1/2} = 1 + 2\left(1 - \frac{1}{2^n}\right) < 3.$$

Step 7 (MCT). (a_n) is increasing (Step 3) and bounded above by 3 (Step 6). By the Monotone Convergence Theorem it converges. Its limit is *defined* to be e , and $2 = a_1 \leq e \leq 3$. $\square$

Example 7.8 (HW3 P4, averages). If (x_n) is increasing, then so is $y_n = \frac{x_1 + \dots + x_n}{n}$. The numerator of $y_{n+1} - y_n$ is $nx_{n+1} - S_n \geq 0$, because each $x_k \leq x_{n+1}$. (Positivity is not actually needed.)

Full proof, step by step. Let $S_n = x_1 + \dots + x_n$, so $y_n = \frac{S_n}{n}$ and $S_{n+1} = S_n + x_{n+1}$.

Step 1 (goal). Show $y_{n+1} - y_n \geq 0$ for every n .

Step 2 (common denominator).

$$y_{n+1} - y_n = \frac{S_{n+1}}{n+1} - \frac{S_n}{n} = \frac{nS_{n+1} - (n+1)S_n}{n(n+1)}.$$

Step 3 (simplify the numerator). Substitute $S_{n+1} = S_n + x_{n+1}$:

$$nS_{n+1} - (n+1)S_n = nS_n + nx_{n+1} - nS_n - S_n = nx_{n+1} - S_n.$$

Step 4 (rewrite as a sum of differences). nx_{n+1} is x_{n+1} added n times, and S_n has n terms, so

$$nx_{n+1} - S_n = \sum_{k=1}^n (x_{n+1} - x_k).$$

Step 5 (each difference is ≥ 0). Since (x_n) is increasing, $x_k \leq x_{k+1} \leq \dots \leq x_{n+1}$ for each $k \leq n$. (Formally, by chaining $x_k \leq x_{k+1}$ repeatedly, or by induction on $n+1-k$.) So every term $x_{n+1} - x_k \geq 0$, and the sum is ≥ 0 .

Step 6 (sign of the fraction). The numerator is ≥ 0 (Step 5) and the denominator $n(n+1) > 0$. So $y_{n+1} - y_n \geq 0$.

Step 7 (conclude). $y_n \leq y_{n+1}$ for all n , so (y_n) is increasing. $\square$

Intuition: adding a new value x_{n+1} that is at least as large as every earlier value cannot lower the average.

8 Subsequences and Cauchy sequences (Week 5)

8.1 Subsequence facts you need

- A **subsequence** is $(x_{n_k})_k$ with $n_1 < n_2 < \dots$. Its index is k , and $n_k \geq k$ **for every k** (Lemma 8.16).
- If $x_n \rightarrow L$ then every subsequence $\rightarrow L$. **So two subsequences with different limits imply divergence**, e.g. $(-1)^n$ with its even and odd terms.
- Every sequence has a monotone subsequence (Lemma 8.17).
- **Bolzano–Weierstrass:** every bounded sequence has a convergent subsequence (Lemma 8.18).

8.2 Definition and intuition

Definition 8.1 (Cauchy sequence). (x_n) is **Cauchy** if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that

$$|x_n - x_m| < \varepsilon \quad \text{for all } n, m \geq N.$$

(The lecture notes write $n, m > N$. The two versions are equivalent: replace N by $N+1$.)

Lecture wording (Week 5, Definition 6) — memorize this.

A sequence $(x_n)_n$ of real numbers is called **Cauchy** if for any $\varepsilon > 0$ there exists N so that $|x_n - x_m| < \varepsilon$ for all $n, m > N$.

In symbols.

$$(x_n) \text{ is Cauchy} \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n, m \geq N : |x_n - x_m| < \varepsilon.$$

Reading the quantifiers in order.

- $\forall \varepsilon > 0$: the challenger names any tolerance, however small.
- $\exists N$: you answer with a cutoff. N **may depend on** ε , but not on n or m .
- $\forall n, m \geq N$: *every* pair of terms past the cutoff, not just neighbors and not just $m = n + 1$.
- $|x_n - x_m| < \varepsilon$: those two terms are within ε of each other.

Side by side with convergence.

$x_n \rightarrow L$	$\forall \varepsilon > 0 \exists N \forall n \geq N : x_n - L < \varepsilon$
(x_n) Cauchy	$\forall \varepsilon > 0 \exists N \forall n, m \geq N : x_n - x_m < \varepsilon$

The only change: the fixed target L is replaced by a second term x_m . That is why the Cauchy condition can be checked without knowing L .

Equivalent form. Writing $m = n + p$: (x_n) is Cauchy iff $\forall \varepsilon > 0 \exists N$ such that $|x_{n+p} - x_n| < \varepsilon$ for all $n \geq N$ and *all* $p \in \mathbb{N}$. The “all p ” is essential. Fixing $p = 1$ gives the weaker condition in the Warning below.

Proof skeleton (copy this shape).

Let $\varepsilon > 0$. Choose $N = \dots$ (*from scratch work*). Let $n, m \geq N$; without loss of generality $m > n$. Then $|x_m - x_n| \leq \dots \leq \dots < \varepsilon$. Hence (x_n) is Cauchy.

Example 8.6 below fills this in for $x_n = \frac{n}{n+1}$.

Intuition. Convergent means “the terms get close to *a number* L .” Cauchy means “the terms get close to *each other*.” **No L appears in the definition.** That is the whole point: you can prove a sequence converges without knowing or guessing its limit, even if it is not monotone (so MCT does not apply).

Negation (you need this to prove “not Cauchy”).

$$(x_n) \text{ is not Cauchy} \iff \exists \varepsilon > 0 \forall N \exists n, m \geq N : |x_n - x_m| \geq \varepsilon.$$

In words: some fixed gap ε keeps appearing between pairs of terms arbitrarily far out.

Warning. Cauchy is *not* the same as $|x_{n+1} - x_n| \rightarrow 0$. The definition controls *all* pairs $n, m \geq N$, not only neighbors. See Example 8.9.

8.3 Methods

Method 8.2 (Proving Cauchy directly).

1. By symmetry assume $m > n$ (if $m = n$ the difference is 0).
2. **Scratch work:** simplify or estimate $|x_m - x_n|$ by an expression that **depends only on n** (not on m) and goes to 0, e.g. $\frac{C}{n}$ or Cr^n .
3. Choose N so that this expression is $< \varepsilon$ when $n \geq N$.
4. Write the forward proof: “Let $\varepsilon > 0$. Choose $N = \dots$. Let $m > n \geq N$. Then $|x_m - x_n| \leq \dots < \varepsilon$.”

Method 8.3 (Telescoping and geometric sums). If $|x_{k+1} - x_k| \leq Cr^k$ for all k , with $0 < r < 1$, then for $m > n$

$$|x_m - x_n| = \left| \sum_{k=n}^{m-1} (x_{k+1} - x_k) \right| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq C \sum_{k=n}^{m-1} r^k < C \frac{r^n}{1-r}.$$

Since $r^n \rightarrow 0$, this is $< \varepsilon$ for large n , so (x_n) is Cauchy. Use this for recursive sequences that oscillate, where MCT fails (Example 8.11).

Method 8.4 (Proving not Cauchy). Find one ε and, for each N , a specific pair $n, m \geq N$ with $|x_n - x_m| \geq \varepsilon$. Good pairs to try: $m = n + 1$ (oscillating sequences), $m = 2n$ (sums and slowly growing sequences).

Method 8.5 (Which tool?). • You know or can guess L : ε - N .

- You can't guess L , but the sequence is monotone and bounded: MCT.
- You can't guess L and the sequence is not monotone: **show it is Cauchy**.
- To show divergence: two subsequences with different limits, or show it is not Cauchy, or show it is unbounded.

8.4 Worked examples

Example 8.6 (A direct Cauchy proof). Prove that $x_n = \frac{n}{n+1}$ is Cauchy.

Scratch work (not part of the proof): $x_n = 1 - \frac{1}{n+1}$. For $m > n$: $|x_m - x_n| = \left| \frac{1}{n+1} - \frac{1}{m+1} \right| = \frac{1}{n+1} - \frac{1}{m+1} < \frac{1}{n+1} < \frac{1}{n}$. This depends only on n . Need $\frac{1}{n} < \varepsilon$, i.e. $n > \frac{1}{\varepsilon}$.

Proof. Let $\varepsilon > 0$. By the Archimedean property choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$. Let $n, m \geq N$; by symmetry assume $m > n$ (if $m = n$ there is nothing to prove). Since $x_k = 1 - \frac{1}{k+1}$,

$$|x_m - x_n| = \frac{1}{n+1} - \frac{1}{m+1} < \frac{1}{n+1} < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

So (x_n) is Cauchy. □

Example 8.7 (Partial sums of $\sum \frac{1}{k^2}$ are Cauchy). Let $s_n = \sum_{k=1}^n \frac{1}{k^2}$. Prove (s_n) is Cauchy.

Scratch work (not part of the proof): For $m > n$, $s_m - s_n = \sum_{k=n+1}^m \frac{1}{k^2}$. We can't sum this exactly, but $\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$ for $k \geq 2$, which telescopes.

Proof. Let $m > n \geq 1$. Every k in the sum satisfies $k \geq n+1 \geq 2$, so $k^2 \geq k(k-1) > 0$ and

$$0 < s_m - s_n = \sum_{k=n+1}^m \frac{1}{k^2} \leq \sum_{k=n+1}^m \left(\frac{1}{k-1} - \frac{1}{k} \right) = \frac{1}{n} - \frac{1}{m} < \frac{1}{n}.$$

Let $\varepsilon > 0$ and choose $N > \frac{1}{\varepsilon}$. For $m > n \geq N$: $|s_m - s_n| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. So (s_n) is Cauchy, and therefore converges, even though we never found its limit. (It is $\frac{\pi^2}{6}$, which you would not be expected to know.) □

Here MCT would also work (s_n is increasing and $s_n < 2$), but the Cauchy proof doesn't need monotonicity, so it also handles $\sum \frac{(-1)^k}{k^2}$ or $\sum \frac{\cos k}{k^2}$ with the same estimate.

Example 8.8 (The harmonic sum is not Cauchy). Let $h_n = \sum_{k=1}^n \frac{1}{k}$. Prove (h_n) is not Cauchy, so it diverges.

Proof. Take $\varepsilon = \frac{1}{2}$. Let $N \in \mathbb{N}$ be arbitrary, and set $n = N$, $m = 2N$, so $n, m \geq N$. The sum $h_{2N} - h_N = \sum_{k=N+1}^{2N} \frac{1}{k}$ has N terms, each $\geq \frac{1}{2N}$ (because $k \leq 2N$). So

$$|h_{2N} - h_N| \geq N \cdot \frac{1}{2N} = \frac{1}{2} = \varepsilon.$$

This is the negation of Cauchy. So (h_n) is not Cauchy and hence not convergent. (Since it is increasing and not convergent, MCT forces it to be unbounded, so in fact $h_n \rightarrow \infty$.) $\square$

Example 8.9 (Neighbors close $\nRightarrow$ Cauchy). Let $x_n = \sqrt{n}$. Then $x_{n+1} - x_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} \rightarrow 0$, but (x_n) is not Cauchy.

Proof. Take $\varepsilon = 1$. Given N , let $n = N$ and $m = 4N$. Then $|x_m - x_n| = 2\sqrt{N} - \sqrt{N} = \sqrt{N} \geq 1 = \varepsilon$. $\square$

Lesson: consecutive differences can shrink while the terms still drift off to ∞ . On a true/false question, “ $x_{n+1} - x_n \rightarrow 0 \Rightarrow$ Cauchy” is **false**.

Example 8.10 ($(-1)^n$ is not Cauchy). Take $\varepsilon = 2$ (or anything ≤ 2). For any N , $n = N$ and $m = N + 1$ give $|x_m - x_n| = |(-1)^{N+1} - (-1)^N| = 2$. So $(-1)^n$ is bounded but not Cauchy, which shows that **bounded $\nRightarrow$ Cauchy**.

Example 8.11 (An oscillating recursive sequence: Cauchy beats MCT). Let $x_1 = 1$ and $x_{n+1} = 1 + \frac{1}{x_n}$. Prove (x_n) converges and find the limit.

Why MCT fails. The terms are 1, 2, $\frac{3}{2}$, $\frac{5}{3}$, $\frac{8}{5}$, $\dots$, which go up, down, up, down. The sequence is not monotone.

Proof. Step 1 ($x_n \geq 1$ for all n). Induction: $x_1 = 1$. If $x_n \geq 1$, then $\frac{1}{x_n} > 0$, so $x_{n+1} = 1 + \frac{1}{x_n} > 1$.

Step 2 (products of neighbors are ≥ 2). For $n \geq 2$, $x_n x_{n-1} = (1 + \frac{1}{x_{n-1}})x_{n-1} = x_{n-1} + 1 \geq 2$ by Step 1.

Step 3 (contraction). For $n \geq 2$,

$$|x_{n+1} - x_n| = \left| \frac{1}{x_n} - \frac{1}{x_{n-1}} \right| = \frac{|x_{n-1} - x_n|}{x_n x_{n-1}} \leq \frac{1}{2} |x_n - x_{n-1}|.$$

Step 4 (geometric bound). Since $|x_2 - x_1| = 1$, induction on Step 3 gives $|x_{k+1} - x_k| \leq (\frac{1}{2})^{k-1}$ for all $k \geq 1$.

Step 5 (Cauchy). For $m > n$, by Method 8.3,

$$|x_m - x_n| \leq \sum_{k=n}^{m-1} \left(\frac{1}{2}\right)^{k-1} = \left(\frac{1}{2}\right)^{n-1} \cdot \frac{1 - (1/2)^{m-n}}{1/2} < \left(\frac{1}{2}\right)^{n-2} = \frac{4}{2^n} \leq \frac{4}{n}$$

(using $2^n \geq n$). Given $\varepsilon > 0$, choose $N > \frac{4}{\varepsilon}$. Then $|x_m - x_n| < \varepsilon$ for $m > n \geq N$. So (x_n) is Cauchy.

Step 6 (find the limit). By the Cauchy criterion, $x_n \rightarrow L$ for some L , and $L \geq 1$ by Step 1 (limits preserve $\geq$). In particular $L \neq 0$, so the quotient rule applies to $x_{n+1} = 1 + \frac{1}{x_n}$. Taking limits on both sides (note $x_{n+1} \rightarrow L$ too) gives $L = 1 + \frac{1}{L}$, i.e. $L^2 - L - 1 = 0$, so $L = \frac{1 \pm \sqrt{5}}{2}$. Since $L \geq 1$, $L = \frac{1 + \sqrt{5}}{2}$. $\square$

Example 8.12 (Sums and products of Cauchy sequences). Let (x_n) and (y_n) be Cauchy. Prove directly from the definition (without the Cauchy criterion) that $(x_n + y_n)$ and $(x_n y_n)$ are Cauchy.

Proof. Sum. Let $\varepsilon > 0$. Choose N_1 with $|x_n - x_m| < \frac{\varepsilon}{2}$ for $n, m \geq N_1$, and N_2 with $|y_n - y_m| < \frac{\varepsilon}{2}$ for $n, m \geq N_2$. Let $N = \max\{N_1, N_2\}$. For $n, m \geq N$,

$$|(x_n + y_n) - (x_m + y_m)| \leq |x_n - x_m| + |y_n - y_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Product. Cauchy sequences are bounded (Lemma 8.15), so there is $M > 0$ with $|x_n| \leq M$ and $|y_n| \leq M$ for all n . Add and subtract $x_n y_m$:

$$|x_n y_n - x_m y_m| = |x_n(y_n - y_m) + y_m(x_n - x_m)| \leq M|y_n - y_m| + M|x_n - x_m|.$$

Let $\varepsilon > 0$ and choose N so that $|x_n - x_m| < \frac{\varepsilon}{2M}$ and $|y_n - y_m| < \frac{\varepsilon}{2M}$ for $n, m \geq N$. Then $|x_n y_n - x_m y_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. $\square$

Example 8.13 (Why completeness matters: Cauchy in $\mathbb{Q}$). Let $q_1 = 1$, $q_2 = 1.4$, $q_3 = 1.41$, $q_4 = 1.414, \dots$ be the decimal truncations of $\sqrt{2}$. Each $q_n \in \mathbb{Q}$ and $|q_m - q_n| < 10^{-(n-1)}$ for $m > n$, so (q_n) is Cauchy. Its limit in $\mathbb{R}$ is $\sqrt{2} \notin \mathbb{Q}$. So **in $\mathbb{Q}$ a Cauchy sequence need not converge**. The proof of the Cauchy criterion below has to use the Completeness Axiom somewhere. It enters through MCT, which needs the existence of $\sup$.

8.5 The long proof: the Cauchy criterion, every step

Theorem 8.14 (Cauchy criterion). A sequence (x_n) of real numbers converges if and only if it is Cauchy.

Proof map. $(\Rightarrow)$ is a short triangle-inequality argument. $(\Leftarrow)$ is built from a chain of lemmas:

$$\begin{aligned} \text{Cauchy} &\xrightarrow{\text{Lem. 8.15}} \text{bounded} \xrightarrow{\text{Lem. 8.17}} \text{bounded monotone subsequence} \\ &\xrightarrow{\text{MCT}} \text{convergent subsequence} \xrightarrow{\text{Lem. 8.19}} \text{convergent}. \end{aligned}$$

Lemmas 8.17 and 8.18 together are Bolzano–Weierstrass. Each lemma below is proved in full.

Part I: convergent $\Rightarrow$ Cauchy

Proof of $(\Rightarrow)$. Step 1 (setup). Suppose $x_n \rightarrow L$. We must show: for every $\varepsilon > 0$ there is N such that $|x_n - x_m| < \varepsilon$ for all $n, m \geq N$.

Step 2 (scratch: how to compare two terms). We know nothing directly about $x_n - x_m$, but we know both terms are close to L . Insert L : $x_n - x_m = (x_n - L) + (L - x_m)$. Each piece will get half of the budget ε .

Step 3 (use convergence with $\frac{\varepsilon}{2}$). Let $\varepsilon > 0$. Then $\frac{\varepsilon}{2} > 0$, so by the definition of $x_n \rightarrow L$ there is $N \in \mathbb{N}$ with

$$|x_k - L| < \frac{\varepsilon}{2} \quad \text{for all } k \geq N.$$

Step 4 (apply it twice). Let $n, m \geq N$. Applying Step 3 with $k = n$ and with $k = m$ gives $|x_n - L| < \frac{\varepsilon}{2}$ and $|x_m - L| < \frac{\varepsilon}{2}$.

Step 5 (triangle inequality).

$$|x_n - x_m| = |(x_n - L) + (L - x_m)| \leq |x_n - L| + |L - x_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Step 6 (conclude). $\varepsilon > 0$ was arbitrary, and the N from Step 3 works for all $n, m \geq N$. So (x_n) is Cauchy. $\square$

Part II: Cauchy $\Rightarrow$ convergent

Lemma 8.15 (Cauchy sequences are bounded). If (x_n) is Cauchy, there is $M > 0$ with $|x_n| \leq M$ for all $n \in \mathbb{N}$.

Proof. Step 1 (pick a specific ε). Boundedness doesn't involve any ε , so we may use the Cauchy property with any fixed tolerance we like. Take $\varepsilon = 1$: there is $N \in \mathbb{N}$ with $|x_n - x_m| < 1$ for all $n, m \geq N$.

Step 2 (anchor at x_N). Fix $m = N$. For every $n \geq N$ we get $|x_n - x_N| < 1$.

Step 3 (bound the tail). For $n \geq N$, by the triangle inequality,

$$|x_n| = |(x_n - x_N) + x_N| \leq |x_n - x_N| + |x_N| < 1 + |x_N|.$$

Step 4 (handle the finitely many early terms). The terms $x_1, \dots, x_{N-1}$ are finitely many real numbers, so they have a largest absolute value. Define

$$M = \max\{|x_1|, |x_2|, \dots, |x_{N-1}|, 1 + |x_N|\}.$$

(If $N = 1$ the list is just $1 + |x_1|$.) Then $M \geq 1 > 0$.

Step 5 (check every n). If $n < N$, then $|x_n|$ is one of the numbers in the max, so $|x_n| \leq M$. If $n \geq N$, then $|x_n| < 1 + |x_N| \leq M$ by Step 3. So $|x_n| \leq M$ for all n . $\square$

Why the max? The Cauchy condition only controls the tail $n \geq N$. A finite list can always be bounded by its maximum, but an infinite list can't. That's why the proof splits into "early" and "tail" terms. The same trick proves "convergent $\Rightarrow$ bounded."

Lemma 8.16 (Indices of a subsequence). If $n_1 < n_2 < \dots$ are natural numbers, then $n_k \geq k$ for all $k \in \mathbb{N}$.

Proof. Induction on k . *Base:* $n_1 \in \mathbb{N}$, so $n_1 \geq 1$. *Step:* assume $n_k \geq k$. Then $n_{k+1} > n_k \geq k$, so $n_{k+1} > k$. Since n_{k+1} and k are integers, $n_{k+1} \geq k + 1$. $\square$

Lemma 8.17 (Monotone Subsequence Theorem). Every sequence (x_n) of real numbers has a monotone subsequence.

Proof. Step 1 (key definition). Call an index $p \in \mathbb{N}$ a **peak** if $x_p \geq x_n$ for every $n > p$. Think of standing at position p and looking right: nothing ahead is higher than x_p .

Step 2 (two exhaustive cases). Either there are infinitely many peaks, or there are finitely many (possibly none). We find a monotone subsequence in each case.

Step 3 (Case 1: infinitely many peaks). List the peaks in increasing order, $p_1 < p_2 < p_3 < \dots$. (Any infinite subset of $\mathbb{N}$ can be listed this way by repeatedly taking the smallest remaining element, using well-ordering.) Set $n_k = p_k$.

- *Subsequence:* $n_1 < n_2 < \dots$ by construction.

- *Monotone:* fix k . Since p_k is a peak, $x_{p_k} \geq x_n$ for every $n > p_k$. In particular $p_{k+1} > p_k$, so $x_{p_k} \geq x_{p_{k+1}}$.

So (x_{n_k}) is **decreasing**.

Step 4 (Case 2: finitely many peaks; setup). Choose $N_0 \in \mathbb{N}$ larger than every peak (if there are no peaks, take $N_0 = 1$). Then **no index $n \geq N_0$ is a peak**.

Step 5 (negate "peak"). " p is a peak" means $\forall n > p : x_p \geq x_n$. So " p is not a peak" means

$$\exists n > p \text{ with } x_n > x_p.$$

Step 6 (recursive construction). Let $n_1 = N_0$. Suppose $n_k \geq N_0$ has been chosen. Then n_k is not a peak, so by Step 5 the set $\{n > n_k : x_n > x_{n_k}\}$ is nonempty. Let n_{k+1} be its smallest element (well-ordering makes the choice definite). Then

$$n_{k+1} > n_k \quad \text{and} \quad x_{n_{k+1}} > x_{n_k}.$$

Also $n_{k+1} > n_k \geq N_0$, so n_{k+1} is again not a peak, and the construction continues. By induction, n_k is defined for every k .

Step 7 (check). The indices are strictly increasing, so (x_{n_k}) is a subsequence, and $x_{n_1} < x_{n_2} < \dots$, so it is (strictly) **increasing**.

Step 8 (conclude). In both cases (x_n) has a monotone subsequence. $\square$

Lemma 8.18 (Bolzano–Weierstrass). Every bounded sequence has a convergent subsequence.

Proof. Step 1. Let $|x_n| \leq M$ for all n . By Lemma 8.17, (x_n) has a monotone subsequence (x_{n_k}) . (Boundedness has not been used yet.)

Step 2 (the subsequence is bounded). Each x_{n_k} is a term of the original sequence, so $|x_{n_k}| \leq M$ for all k .

Step 3 (MCT). (x_{n_k}) is monotone and bounded, so by the Monotone Convergence Theorem it converges. **This is where completeness enters:** MCT says the limit is the sup (or inf) of the terms, and that sup exists by the Completeness Axiom. $\square$

Lemma 8.19 (A Cauchy sequence follows its convergent subsequence). If (x_n) is Cauchy and some subsequence $x_{n_k} \rightarrow L$, then $x_n \rightarrow L$.

Proof. Step 1 (goal). Let $\varepsilon > 0$. We need N with $|x_n - L| < \varepsilon$ for all $n \geq N$.

Step 2 (scratch: insert a subsequence term). We control $|x_n - x_m|$ (Cauchy) and $|x_{n_k} - L|$ (subsequence). Insert x_{n_k} : $|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L|$, and give each piece $\frac{\varepsilon}{2}$.

Step 3 (Cauchy with $\frac{\varepsilon}{2}$). There is N_1 with $|x_n - x_m| < \frac{\varepsilon}{2}$ for all $n, m \geq N_1$.

Step 4 (subsequence with $\frac{\varepsilon}{2}$). There is K with $|x_{n_k} - L| < \frac{\varepsilon}{2}$ for all $k \geq K$.

Step 5 (pick one good subsequence index). Let $k_0 = \max\{K, N_1\}$. Then $k_0 \geq K$, so $|x_{n_{k_0}} - L| < \frac{\varepsilon}{2}$. By Lemma 8.16, $n_{k_0} \geq k_0 \geq N_1$. Note that k_0 depends only on ε , not on n .

Step 6 (choose N and verify). Let $N = N_1$ and let $n \geq N$. Both n and n_{k_0} are $\geq N_1$, so Step 3 with $m = n_{k_0}$ gives $|x_n - x_{n_{k_0}}| < \frac{\varepsilon}{2}$. Therefore

$$|x_n - L| \leq |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Step 7. ε was arbitrary, so $x_n \rightarrow L$. $\square$

Proof of Theorem 8.14, ($\Leftarrow$). Let (x_n) be Cauchy.

1. By Lemma 8.15, (x_n) is bounded.

2. By Bolzano–Weierstrass (Lemma 8.18), it has a convergent subsequence $x_{n_k} \rightarrow L$ for some $L \in \mathbb{R}$.

3. By Lemma 8.19, $x_n \rightarrow L$.

Together with Part I, this proves Theorem 8.14. $\square$

Exam-length version (what to write in about 10 minutes)

If you are asked to “prove every Cauchy sequence converges” and may quote B–W:

Proof. Let (x_n) be Cauchy. With $\varepsilon = 1$ there is N such that $|x_n - x_N| < 1$ for $n \geq N$, so $|x_n| \leq \max\{|x_1|, \dots, |x_{N-1}|, 1 + |x_N|\}$ for all n , and (x_n) is bounded. By Bolzano–Weierstrass some subsequence $x_{n_k} \rightarrow L$. Let $\varepsilon > 0$. Choose N_1 with $|x_n - x_m| < \frac{\varepsilon}{2}$ for $n, m \geq N_1$, and K with $|x_{n_k} - L| < \frac{\varepsilon}{2}$ for $k \geq K$. Fix $k = \max\{K, N_1\}$, so $n_k \geq k \geq N_1$. For $n \geq N_1$, $|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L| < \varepsilon$. So $x_n \rightarrow L$. $\square$

Points graders look for: (1) boundedness proved, with the finite-max trick; (2) B–W named; (3) the $\frac{\varepsilon}{2}$ split; (4) the justification $n_k \geq k \geq N_1$.

8.6 The Nested Interval Theorem

Theorem 8.20 (Nested Interval Theorem). Let $I_n = [a_n, b_n]$ be **closed, bounded** intervals that are **nested**, i.e. $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$. Then

- (a) $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$; in fact $\bigcap_n I_n = [\alpha, \beta]$ where $\alpha = \sup_n a_n$ and $\beta = \inf_n b_n$;
- (b) if in addition $b_n - a_n \rightarrow 0$, then $\bigcap_n I_n$ is a single point $\{\alpha\}$, and $a_n \rightarrow \alpha$, $b_n \rightarrow \alpha$.

Picture. The left endpoints move right, the right endpoints move left, and they never cross. Completeness says there is no “hole” where they could meet, so some real number is trapped in all of the intervals.

Full proof, step by step. Step 1 (unpack “nested”). $I_{n+1} \subseteq I_n$ means $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$, i.e.

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n \quad \text{for all } n.$$

So (a_n) is increasing and (b_n) is decreasing.

Step 2 (left endpoints never pass right endpoints). Claim: $a_m \leq b_n$ for all $m, n \in \mathbb{N}$ (not only $m = n$).

- If $m \leq n$: $a_m \leq a_n \leq b_n$ (a increasing, then $a_n \leq b_n$).
- If $m > n$: $a_m \leq b_m \leq b_n$ ($a_m \leq b_m$, then b decreasing).

Step 3 (define α by completeness). Let $A = \{a_m : m \in \mathbb{N}\}$. It is nonempty, and by Step 2 (with $n = 1$) b_1 is an upper bound. By the Completeness Axiom, $\alpha = \sup A$ exists.

Step 4 ($\alpha \in I_n$ for every n). Fix n .

- α is an upper bound of A , so $a_n \leq \alpha$.
- By Step 2, b_n is an upper bound of A . Since α is the *least* upper bound, $\alpha \leq b_n$.

So $a_n \leq \alpha \leq b_n$, i.e. $\alpha \in I_n$. Since n was arbitrary, $\alpha \in \bigcap_n I_n$, and the intersection is nonempty.

Step 5 (the whole intersection). Similarly $\beta = \inf\{b_n\}$ exists, and Step 4 shows α is a lower bound of $\{b_n\}$, so $\alpha \leq \beta$. If $x \in [\alpha, \beta]$, then $a_n \leq \alpha \leq x \leq \beta \leq b_n$ for every n , so $x \in \bigcap I_n$. Conversely, if $x \in \bigcap I_n$ then $a_n \leq x \leq b_n$ for all n , so x is an upper bound of $\{a_n\}$ (giving $\alpha \leq x$) and a lower bound of $\{b_n\}$ (giving $x \leq \beta$). Hence $\bigcap I_n = [\alpha, \beta]$.

Step 6 (part (b): uniqueness). Suppose $b_n - a_n \rightarrow 0$ and $x, y \in \bigcap I_n$. For every n , both x and y lie in $[a_n, b_n]$, so $|x - y| \leq b_n - a_n$. Let $\varepsilon > 0$. For large n , $b_n - a_n < \varepsilon$, so $|x - y| < \varepsilon$. Since ε was arbitrary, $|x - y| = 0$, so $x = y$ (the “ $< \varepsilon$ for all ε ” trick). So $\bigcap I_n = \{\alpha\}$.

Step 7 (endpoints converge). $0 \leq \alpha - a_n \leq b_n - a_n \rightarrow 0$ and $0 \leq b_n - \alpha \leq b_n - a_n \rightarrow 0$, so by the Squeeze Theorem $a_n \rightarrow \alpha$ and $b_n \rightarrow \alpha$. $\square$

Short proof via MCT. (a_n) is increasing and bounded above by b_1 , so $a_n \rightarrow \alpha$ by MCT. For fixed n , $a_m \leq b_n$ for all m , and limits preserve $\leq$, so $\alpha \leq b_n$. Also $a_n \leq \alpha$ because an increasing sequence stays below its limit. So $\alpha \in I_n$ for all n .

Example 8.21 (Every hypothesis is needed).

- **Closed is needed:** $I_n = (0, \frac{1}{n})$ are nested, bounded and open, but $\bigcap_n (0, \frac{1}{n}) = \emptyset$. If $x > 0$ were in all of them, pick $n > \frac{1}{x}$ (Archimedean). Then $x \geq \frac{1}{n}$, so $x \notin I_n$. And $x \leq 0$ is in none of them.
- **Bounded is needed:** $I_n = [n, \infty)$ are nested and closed, but $\bigcap_n I_n = \emptyset$: any x fails to be in I_n once $n > x$ (Archimedean).
- **Nested is needed:** $[n, n+1]$ are closed and bounded but not nested, and no point is in all of them.
- **$\mathbb{R}$ (completeness) is needed:** in $\mathbb{Q}$, let I_n be the rational points of $[q_n, q_n + 10^{-(n-1)}]$, where q_n is the decimal truncation of $\sqrt{2}$ to $n-1$ decimal places. Their intersection in $\mathbb{R}$ is $\{\sqrt{2}\}$, so in $\mathbb{Q}$ it is empty.

Example 8.22 (Bisection proof of Bolzano–Weierstrass using NIT). Let $|x_n| \leq M$ for all n . We find a convergent subsequence without using Lemma 8.17.

Proof. Step 1 (start). Let $I_1 = [-M, M]$. It contains x_n for *infinitely many* n (in fact all n).

Step 2 (halve and keep an infinite half). Suppose I_k has been chosen, has length $\frac{2M}{2^{k-1}}$, and contains x_n for infinitely many n . Split it into its closed left and right halves. If each half contained x_n for only finitely many n , then I_k would too. So at least one half contains x_n for infinitely many n . Call it I_{k+1} (choose the left half if both work). Its length is $\frac{2M}{2^k}$, and $I_{k+1} \subseteq I_k$.

Step 3 (pick indices). Let $n_1 = 1$ (so $x_{n_1} \in I_1$). Given n_k , the set $\{n : x_n \in I_{k+1}\}$ is infinite, so it contains some $n > n_k$. Let n_{k+1} be the smallest such n . Then $n_1 < n_2 < \dots$ and $x_{n_k} \in I_k$ for all k .

Step 4 (NIT). The I_k are closed, bounded and nested, and their lengths $\frac{4M}{2^k} \rightarrow 0$. By Theorem 8.20 there is a single $L \in \bigcap_k I_k$.

Step 5 (convergence). For each k , both x_{n_k} and L lie in I_k , so $|x_{n_k} - L| \leq \frac{4M}{2^k} \leq \frac{4M}{k}$ (using $2^k \geq k$). Given $\varepsilon > 0$, choose $K > \frac{4M}{\varepsilon}$. For $k \geq K$, $|x_{n_k} - L| < \varepsilon$. So $x_{n_k} \rightarrow L$. $\square$

Compare. The peak proof uses completeness through MCT. The bisection proof uses it through NIT. MCT, NIT, B–W and the Cauchy criterion are all ways of saying that $\mathbb{R}$ has no holes.

Example 8.23 (Using NIT to locate a point: $\sqrt{2}$ by bisection). Let $I_1 = [1, 2]$. At each step keep the half $[a, b]$ with $a^2 \leq 2 \leq b^2$: $[1, 1.5]$, $[1.25, 1.5]$, $[1.375, 1.5]$, $[1.375, 1.4375]$, $\dots$. The intervals are nested and closed, with lengths $2^{-(n-1)} \rightarrow 0$, so NIT gives a unique c in all of them. Since $a_n^2 \leq 2 \leq b_n^2$ and $a_n, b_n \rightarrow c$, the product rule gives $a_n^2, b_n^2 \rightarrow c^2$. Limits preserve $\leq$, so $c^2 \leq 2 \leq c^2$, i.e. $c^2 = 2$. This is another proof that $\sqrt{2}$ exists in $\mathbb{R}$, and it is the same idea as the bisection method for root-finding.

9 Common mistakes that cost points

1. **Writing ε – N proofs backwards.** Scratch work goes from the goal to N . The proof must go from “Let $\varepsilon > 0$ ” to “ $< \varepsilon$ ”.
2. **Choosing N that depends on n .** N may depend only on ε (and fixed constants).
3. **Forgetting to show that the sup is an upper bound.** Both parts (i) and (ii) are needed.

4. **Assuming the sup is in the set.** $\sup(0, 1) = 1 \notin (0, 1)$.
5. **Using limit laws when a limit doesn't exist**, e.g. splitting $\frac{\sin n}{n}$ as $\lim \sin n \cdot \lim \frac{1}{n}$.
6. **Passing to the limit before proving convergence** in recursive sequences. MCT comes first, then solve $L = f(L)$.
7. **Multiplying an inequality by something that might be negative.**
8. **Induction without writing $P(n + 1)$ first**, and then “proving” the wrong target.
9. **Strict vs. non-strict:** limits preserve $\leq$ but not $<$.
10. **Cauchy estimate that depends on m .** The bound on $|x_m - x_n|$ must depend only on n (or on N), e.g. $\frac{1}{n} - \frac{1}{m} < \frac{1}{n}$. Drop the m -term before choosing N .
11. **Checking only neighbors.** Showing $|x_{n+1} - x_n| \rightarrow 0$ does *not* prove Cauchy (Example 8.9).
12. **Putting L into a Cauchy proof.** If you are asked to show a sequence is Cauchy because you *don't* know L , you may not use L .
13. **NIT with open or unbounded intervals.** The theorem needs closed *and* bounded intervals (Example 8.21).

10 Practice problems

Try these under exam conditions. Solutions are in the next section. Problems 27–34 are on Cauchy sequences and Problems 35–37 are on nested intervals.

Practice Problem 1. Prove by induction: $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for all $n \in \mathbb{N}$.

Practice Problem 2. Prove by induction: $2^n \geq n + 1$ for all $n \in \mathbb{N}$.

Practice Problem 3. Prove that $\sqrt{3}$ is irrational.

Practice Problem 4. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Practice Problem 5. Find sup and inf of $S = \{(-1)^n (1 - \frac{1}{n}) : n \in \mathbb{N}\}$, say whether each is attained, and prove your claim about the sup.

Practice Problem 6. Find sup and inf of $T = \{\frac{n}{n+1} : n \in \mathbb{N}\}$ with proof.

Practice Problem 7. Let A be nonempty and bounded above, and $c > 0$. Prove $\sup(cA) = c\sup A$.

Practice Problem 8. Let $A \subseteq B$ be nonempty and bounded. Prove $\inf B \leq \inf A \leq \sup A \leq \sup B$.

Practice Problem 9. Use the definition to prove $\lim_{n \rightarrow \infty} \frac{2n^2 + 1}{n^2 + 3} = 2$.

Practice Problem 10. Use the definition to prove $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

Practice Problem 11. Prove $\lim_{n \rightarrow \infty} \frac{n^2}{n+1} = \infty$ from the definition.

Practice Problem 12. Prove that if $x_n \rightarrow L$ then $|x_n| \rightarrow |L|$. Is the converse true?

Practice Problem 13. Prove that $x_n = \frac{(-1)^n n}{n+1}$ diverges.

Practice Problem 14. Prove: if $x_n \rightarrow L$ and $x_n \geq 0$ for all n , then $L \geq 0$.

Practice Problem 15. Let $x_1 = 2$ and $x_{n+1} = \frac{x_n+3}{2}$. Prove that (x_n) converges and find its limit.

Practice Problem 16. Compute (limit laws allowed): (a) $\lim \frac{5n^3-n}{2n^3+7}$ (b) $\lim \frac{n+1}{n^2+1}$ (c) $\lim (\frac{2}{3})^n$
(d) $\lim \sum_{k=0}^n \frac{1}{2^k}$ (e) $\lim \frac{\cos(n^2)}{\sqrt{n}}$.

Practice Problem 17 (True or false: justify or give a counterexample).

- (a) If $x_n > 0$ for all n and $x_n \rightarrow L$, then $L > 0$.
- (b) Every bounded sequence converges.
- (c) Every convergent sequence is bounded.
- (d) If $x_n y_n \rightarrow 0$ then $x_n \rightarrow 0$ or $y_n \rightarrow 0$.
- (e) If (x_n) and (y_n) both diverge, then $(x_n + y_n)$ diverges.
- (f) Every nonempty subset of $\mathbb{Q}$ that is bounded above has a sup in $\mathbb{Q}$.
- (g) If $\sup S$ exists and $\sup S \in S$, then S has a maximum.

Practice Problem 18. Let A, B be nonempty and bounded above. Prove $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

Practice Problem 19. Find sup and inf of $\{\frac{1}{n} + \frac{1}{m} : n, m \in \mathbb{N}\}$.

Practice Problem 20. Prove that if $x_n \rightarrow 0$ and (y_n) is bounded, then $x_n y_n \rightarrow 0$. Use it to find $\lim \frac{(-1)^n n}{n^2 + 1}$.

Practice Problem 21. Prove that if $x_n \rightarrow \infty$ then $\frac{1}{x_n} \rightarrow 0$ (from the definitions).

Practice Problem 22. Use the definition to prove $\lim_{n \rightarrow \infty} \frac{3n + 1}{n - 2} = 3$ (consider $n \geq 3$).

Practice Problem 23. Prove that $\frac{n}{2^n} \rightarrow 0$.

Practice Problem 24. Solve $|2x - 1| < |x + 2|$.

Practice Problem 25. Prove: if $x_n \rightarrow L > 0$, then $x_n > \frac{L}{2}$ for all sufficiently large n .

Practice Problem 26. Show that $S = \{x \in \mathbb{R} : x^2 < 5\}$ is bounded above, explain why $\sup S$ exists without knowing its value, and identify $\sup S$.

Cauchy sequences

Practice Problem 27. Prove directly from the definition that $x_n = \frac{3n + 2}{n + 1}$ is Cauchy.

Practice Problem 28. Let $s_n = \sum_{k=0}^n \frac{1}{k!}$. Prove (s_n) is Cauchy. (Hint: first show $k! \geq 2^{k-1}$ for $k \geq 1$.)

Practice Problem 29. Let $t_n = \sum_{k=1}^n \frac{1}{\sqrt{k}}$. Prove (t_n) is not Cauchy.

Practice Problem 30. Without looking at Lemma 8.19: prove that if (x_n) is Cauchy and a subsequence $x_{n_k} \rightarrow L$, then $x_n \rightarrow L$.

Practice Problem 31. Suppose $0 < r < 1$ and $|x_{n+1} - x_n| \leq r^n$ for all n . Prove (x_n) is Cauchy (and hence converges).

Practice Problem 32. Let $x_1 = 1$, $x_2 = 2$, and $x_{n+2} = \frac{x_n + x_{n+1}}{2}$. Prove (x_n) is Cauchy and find its limit. (Hint: look at $d_n = x_{n+1} - x_n$, and at $x_{n+1} + \frac{1}{2}x_n$.)

Practice Problem 33 (True or false: justify or give a counterexample).

- (a) If $x_{n+1} - x_n \rightarrow 0$, then (x_n) is Cauchy.
- (b) Every Cauchy sequence of rational numbers has a rational limit.
- (c) Every subsequence of a Cauchy sequence is Cauchy.
- (d) If (x_n) is Cauchy, then $(|x_n|)$ is Cauchy. Is the converse true?
- (e) Every bounded sequence is Cauchy.
- (f) If (x_n) is Cauchy and $a \leq x_n \leq b$ for all n , then $\lim x_n \in [a, b]$.

Practice Problem 34. Prove from the definition that every Cauchy sequence is bounded. Then give a bounded sequence that is not Cauchy.

Nested intervals

Practice Problem 35. Find $\bigcap_n [-\frac{1}{n}, \frac{1}{n}]$ and $\bigcap_n (0, \frac{1}{n}]$, with proof. Why doesn't the second one contradict the Nested Interval Theorem?

Practice Problem 36. Let $I_n = [1 - \frac{1}{n}, 1 + \frac{1}{n^2}]$. Show the intervals are nested and find $\bigcap_n I_n$.

Practice Problem 37. Let $I_n = [a_n, b_n]$ be closed, nested, with $b_n - a_n \rightarrow 0$, and let α be the point in $\bigcap_n I_n$. If $x_n \in I_n$ for every n , prove $x_n \rightarrow \alpha$.

11 Solutions

1. Base case: $1 = 1^2$. Target $P(n+1)$: $1 + 3 + \cdots + (2n-1) + (2n+1) = (n+1)^2$. By the hypothesis the left side is $n^2 + 2n + 1 = (n+1)^2$. $\square$

2. Base case: $2 \geq 2$. Assume $2^n \geq n+1$. Then $2^{n+1} = 2 \cdot 2^n \geq 2(n+1) = 2n+2 \geq n+2$. $\square$

3. Suppose $\sqrt{3} = p/q$ in lowest terms, so $p^2 = 3q^2$. If $3 \nmid p$, then $p = 3k \pm 1$ and $p^2 = 9k^2 \pm 6k + 1 = 3(3k^2 \pm 2k) + 1$ is not a multiple of 3, a contradiction. So $p = 3k$, which gives $9k^2 = 3q^2$, $q^2 = 3k^2$, and so $3 \mid q$. This contradicts lowest terms. $\square$

4. ($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$, and $x \in B$ or $x \in C$. Case $x \in B$: $x \in A \cap B$. Case $x \in C$: $x \in A \cap C$. Either way x is in the union. ($\supseteq$) If $x \in A \cap B$, then $x \in A$ and $x \in B \subseteq B \cup C$. The case $x \in A \cap C$ is the same. $\square$

5. The elements are $0, \frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots$. Even n gives $1 - \frac{1}{n} \in [\frac{1}{2}, 1)$. Odd n gives $-(1 - \frac{1}{n}) \in (-1, 0]$. So $\sup S = 1$ and $\inf S = -1$, and *neither is attained*.

Proof that $\sup S = 1$: every element satisfies $(-1)^n(1 - \frac{1}{n}) \leq 1 - \frac{1}{n} < 1$, so 1 is an upper bound. Given $\varepsilon > 0$, choose an even n with $\frac{1}{n} < \varepsilon$ (Archimedean; if n is odd, use $n+1$). Then $1 - \frac{1}{n} \in S$ and $1 - \frac{1}{n} > 1 - \varepsilon$. $\square$ (The inf is symmetric, using odd n .)

6. $\frac{n}{n+1} = 1 - \frac{1}{n+1}$, which is increasing. So $\inf T = \min T = \frac{1}{2}$ (at $n = 1$), and $\sup T = 1$ is not attained. Upper bound: $\frac{n}{n+1} < 1$. Given $\varepsilon > 0$, pick n with $\frac{1}{n+1} < \varepsilon$. Then $\frac{n}{n+1} > 1 - \varepsilon$. $\square$

7. Let $\alpha = \sup A$. For $a \in A$: $a \leq \alpha$, so $ca \leq c\alpha$ (as $c > 0$), and $c\alpha$ is an upper bound of cA . Given $\varepsilon > 0$, $\frac{\varepsilon}{c} > 0$, so there is $a \in A$ with $a > \alpha - \frac{\varepsilon}{c}$. Then $ca \in cA$ and $ca > c\alpha - \varepsilon$. By Theorem 3.3, $\sup(cA) = c\alpha$. $\square$

8. Middle: pick any $a \in A$, so $\inf A \leq a \leq \sup A$. Right: $\sup B$ is an upper bound of $B \supseteq A$, so it is an upper bound of A . Hence $\sup A \leq \sup B$ ($\sup$ is the *least* upper bound). Left: the same argument with lower bounds. $\square$

9. $\left| \frac{2n^2+1}{n^2+3} - 2 \right| = \left| \frac{2n^2+1-2n^2-6}{n^2+3} \right| = \frac{5}{n^2+3} < \frac{5}{n^2} \leq \frac{5}{n}$. Let $\varepsilon > 0$ and choose $N > \frac{5}{\varepsilon}$. For $n \geq N$: $\left| \frac{2n^2+1}{n^2+3} - 2 \right| < \frac{5}{n} \leq \frac{5}{N} < \varepsilon$. $\square$

10. $\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{\sqrt{n}}$. We want $\frac{1}{\sqrt{n}} < \varepsilon \iff n > \frac{1}{\varepsilon^2}$. Let $\varepsilon > 0$ and choose $N > \frac{1}{\varepsilon^2}$. For $n \geq N$: $|\sqrt{n+1} - \sqrt{n} - 0| < \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \varepsilon$. $\square$

11. For $n \geq 1$, $n+1 \leq 2n$, so $\frac{n^2}{n+1} \geq \frac{n^2}{2n} = \frac{n}{2}$. Given M , choose $N > 2M$. For $n \geq N$: $\frac{n^2}{n+1} \geq \frac{n}{2} \geq \frac{N}{2} > M$. $\square$

12. By the reverse triangle inequality, $||x_n| - |L|| \leq |x_n - L| < \varepsilon$ for $n \geq N$, with the same N that works for $x_n \rightarrow L$. The converse is *false*: $x_n = (-1)^n$ has $|x_n| \rightarrow 1$ but diverges. (It is true when $L = 0$, since $|x_n - 0| = ||x_n| - 0|$.) $\square$

13. Suppose $x_n \rightarrow L$. The even terms are $\frac{n}{n+1} \rightarrow 1$ and the odd terms are $-\frac{n}{n+1} \rightarrow -1$. Directly: take $\varepsilon = \frac{1}{2}$. For $n \geq N$ large, even n gives $x_n > \frac{1}{2}$ and odd n gives $x_n < -\frac{1}{2}$, so these terms are more than 1 apart. But $|x_n - x_m| \leq |x_n - L| + |L - x_m| < 1$, a contradiction. (Or: two subsequences with different limits.) $\square$

14. Suppose $L < 0$. Take $\varepsilon = -\frac{L}{2} > 0$. For large n , $x_n < L + \varepsilon = \frac{L}{2} < 0$, which contradicts $x_n \geq 0$. $\square$

15. The terms are $x_1 = 2$, $x_2 = 2.5$, $x_3 = 2.75$, so the guess is increasing with limit 3. *Bound* $x_n < 3$: true for $n = 1$; if $x_n < 3$ then $x_{n+1} < \frac{3+x_n}{2} = 3$. *Increasing*: $x_{n+1} - x_n = \frac{3-x_n}{2} > 0$ by the bound. By MCT, $x_n \rightarrow L$. Taking limits in $x_{n+1} = \frac{x_n+3}{2}$ gives $L = \frac{L+3}{2}$, so $L = 3$. $\square$

16. (a) $\frac{5}{2}$ (b) 0 (lower degree on top) (c) 0 since $|\frac{2}{3}| < 1$ (d) $\frac{1}{1-1/2} = 2$ (e) 0, since $\left| \frac{\cos(n^2)}{\sqrt{n}} \right| \leq \frac{1}{\sqrt{n}} \rightarrow 0$ (squeeze).

17.

- (a) False: $\frac{1}{n} > 0$ but $\frac{1}{n} \rightarrow 0$. (You only get $L \geq 0$.)
- (b) False: $(-1)^n$. (It *does* have a convergent subsequence, by B-W.)
- (c) True: the “convergent $\Rightarrow$ bounded” theorem.
- (d) False: $x_n = 1, 0, 1, 0, \dots$ and $y_n = 0, 1, 0, 1, \dots$ have $x_n y_n = 0$, but neither converges to 0.
- (e) False: $x_n = (-1)^n$ and $y_n = -(-1)^n$ sum to 0.
- (f) False: $\{q \in \mathbb{Q} : q^2 < 2\}$ has no sup in $\mathbb{Q}$, since the sup would be $\sqrt{2}$. This is exactly why $\mathbb{R}$ needs the completeness axiom.
- (g) True: $\sup S$ is an upper bound in S , which is the definition of a maximum.

18. Let $\alpha = \sup A$, $\beta = \sup B$, $M = \max\{\alpha, \beta\}$. *Upper bound*: if $x \in A \cup B$, then $x \leq \alpha \leq M$ or $x \leq \beta \leq M$. *Least*: say $M = \alpha$. Given $\varepsilon > 0$, there is $a \in A \subseteq A \cup B$ with $a > \alpha - \varepsilon = M - \varepsilon$. $\square$

19. $\sup = \max = 2$ (at $n = m = 1$), since $\frac{1}{n} + \frac{1}{m} \leq 1 + 1$. $\inf = 0$, not attained: 0 is a lower bound since every element is > 0 . Given $\varepsilon > 0$, pick n with $\frac{1}{n} < \frac{\varepsilon}{2}$. Then $\frac{1}{n} + \frac{1}{n} < \varepsilon$, with $m = n$. $\square$

20. Let $|y_n| \leq M$ with $M > 0$. Given $\varepsilon > 0$, choose N with $|x_n| < \frac{\varepsilon}{M}$ for $n \geq N$. Then $|x_n y_n| \leq M|x_n| < \varepsilon$. Application: $\frac{(-1)^n n}{n^2+1} = (-1)^n \cdot \frac{n}{n^2+1}$, where $(-1)^n$ is bounded and $0 < \frac{n}{n^2+1} < \frac{1}{n} \rightarrow 0$. So the limit is 0. $\square$

21. Let $\varepsilon > 0$. Apply $x_n \rightarrow \infty$ with $M = \frac{1}{\varepsilon}$: there is N with $x_n > \frac{1}{\varepsilon} > 0$ for $n \geq N$. Then $0 < \frac{1}{x_n} < \varepsilon$, so $\left| \frac{1}{x_n} - 0 \right| < \varepsilon$. $\square$

22. $\frac{3n+1}{n-2} - 3 = \frac{7}{n-2} > 0$ for $n \geq 3$. For $n \geq 3$ we have $n-2 \geq \frac{n}{3}$ (equivalent to $\frac{2n}{3} \geq 2$), so $\frac{7}{n-2} \leq \frac{21}{n}$. Given $\varepsilon > 0$, choose $N > \max\{3, \frac{21}{\varepsilon}\}$. For $n \geq N$: $\left| \frac{3n+1}{n-2} - 3 \right| \leq \frac{21}{n} \leq \frac{21}{N} < \varepsilon$. $\square$

23. By the binomial theorem, $2^n \geq \binom{n}{2} = \frac{n(n-1)}{2}$ for $n \geq 2$, so $0 < \frac{n}{2^n} \leq \frac{2}{n-1}$. Given $\varepsilon > 0$, choose $N > 1 + \frac{2}{\varepsilon}$. For $n \geq N$: $\frac{n}{2^n} \leq \frac{2}{n-1} < \varepsilon$. $\square$

24. Both sides are ≥ 0 , so the inequality is equivalent to comparing squares: $(2x - 1)^2 < (x + 2)^2 \iff 3x^2 - 8x - 3 < 0 \iff (3x + 1)(x - 3) < 0 \iff -\frac{1}{3} < x < 3$. (Alternatively, split into cases at $x = \frac{1}{2}$ and $x = -2$.)

25. Take $\varepsilon = \frac{L}{2} > 0$. There is N with $|x_n - L| < \frac{L}{2}$ for $n \geq N$, so $x_n > L - \frac{L}{2} = \frac{L}{2}$. **This is the “eventually positive” lemma, used in the quotient rule.** $\square$

26. If $x \geq 3$ then $x^2 \geq 9 > 5$, so every $x \in S$ satisfies $x < 3$, and 3 is an upper bound. $S \neq \emptyset$ since $0 \in S$. By the Completeness Axiom, $\sup S$ exists, even though we haven’t computed it. In fact $\sup S = \sqrt{5}$, whose existence was proved in lecture by exactly this sup argument for $\sqrt{2}$.

Cauchy sequences

27. $x_n = 3 - \frac{1}{n+1}$. For $m > n$: $|x_m - x_n| = \frac{1}{n+1} - \frac{1}{m+1} < \frac{1}{n}$. Let $\varepsilon > 0$ and choose $N > \frac{1}{\varepsilon}$. For $m > n \geq N$: $|x_m - x_n| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. $\square$

28. *Claim:* $k! \geq 2^{k-1}$ for $k \geq 1$. Base: $1! = 1 = 2^0$. Step: $(k+1)! = (k+1)k! \geq 2 \cdot 2^{k-1} = 2^k$. For $m > n \geq 1$:

$$0 < s_m - s_n = \sum_{k=n+1}^m \frac{1}{k!} \leq \sum_{k=n+1}^m \frac{1}{2^{k-1}} < \frac{1/2^n}{1 - \frac{1}{2}} = \frac{1}{2^{n-1}} \leq \frac{1}{n},$$

using $2^{n-1} \geq n$ (Problem 2). Given $\varepsilon > 0$, choose $N > \frac{1}{\varepsilon}$. Then $|s_m - s_n| < \varepsilon$ for $m > n \geq N$. $\square$ (The limit is e .)

29. Take $\varepsilon = \frac{1}{2}$. Given N , let $n = N$ and $m = 2N$. The sum $t_{2N} - t_N = \sum_{k=N+1}^{2N} \frac{1}{\sqrt{k}}$ has N terms, each $\geq \frac{1}{\sqrt{2N}}$, so $t_{2N} - t_N \geq \frac{N}{\sqrt{2N}} = \sqrt{\frac{N}{2}} \geq \frac{1}{\sqrt{2}} > \frac{1}{2}$. $\square$

30. See Lemma 8.19. The key line: choose N_1 (Cauchy, $\frac{\varepsilon}{2}$) and K (subsequence, $\frac{\varepsilon}{2}$), fix $k = \max\{K, N_1\}$ so $n_k \geq k \geq N_1$. Then for $n \geq N_1$, $|x_n - L| \leq |x_n - x_{n_k}| + |x_{n_k} - L| < \varepsilon$. $\square$

31. For $m > n$, telescoping and the triangle inequality give $|x_m - x_n| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq \sum_{k=n}^{m-1} r^k < \frac{r^n}{1-r}$. Since $r^n \rightarrow 0$ (basic limit, $0 < r < 1$), given $\varepsilon > 0$ there is N with $r^N < \varepsilon(1-r)$. For $m > n \geq N$, $r^n \leq r^N$, so $|x_m - x_n| < \frac{r^N}{1-r} < \varepsilon$. $\square$

32. Let $d_n = x_{n+1} - x_n$. Then $d_{n+1} = \frac{x_n + x_{n+1}}{2} - x_{n+1} = -\frac{1}{2}d_n$, and $d_1 = 1$, so $|d_n| = \frac{1}{2^{n-1}} = 2\left(\frac{1}{2}\right)^n$. As in Problem 31 (with an extra factor 2), $|x_m - x_n| < \frac{2(1/2)^n}{1/2} = \frac{4}{2^n} \leq \frac{4}{n}$ for $m > n$, so (x_n) is Cauchy and converges to some L .

Finding L . Taking limits in the recurrence gives $L = \frac{L+L}{2}$, which is true for every L and tells us nothing. Use the invariant instead: $x_{n+2} + \frac{1}{2}x_{n+1} = \frac{x_n + x_{n+1}}{2} + \frac{1}{2}x_{n+1} = x_{n+1} + \frac{1}{2}x_n$, so $x_{n+1} + \frac{1}{2}x_n = x_2 + \frac{1}{2}x_1 = \frac{5}{2}$ for all n . Letting $n \rightarrow \infty$: $L + \frac{L}{2} = \frac{5}{2}$, so $L = \frac{5}{3}$. $\square$

33.

- (a) False: $x_n = \sqrt{n}$ (Example 8.9).
- (b) False: the decimal truncations of $\sqrt{2}$ are rational and Cauchy, but their limit is $\sqrt{2} \notin \mathbb{Q}$.
- (c) True: if $|x_n - x_m| < \varepsilon$ for $n, m \geq N$, then for $k, j \geq N$ we have $n_k \geq k \geq N$ and $n_j \geq N$, so $|x_{n_k} - x_{n_j}| < \varepsilon$.
- (d) True: $||x_n| - |x_m|| \leq |x_n - x_m|$ (reverse triangle inequality), with the same N . The converse is false: $(-1)^n$.
- (e) False: $(-1)^n$.
- (f) True: it converges (Cauchy criterion), and limits preserve $a \leq x_n \leq b$.

34. See Lemma 8.15: use $\varepsilon = 1$ to get $|x_n| < 1 + |x_N|$ for $n \geq N$, then $M = \max\{|x_1|, \dots, |x_{N-1}|, 1 + |x_N|\}$. A bounded sequence that is not Cauchy: $(-1)^n$. $\square$

Nested intervals

35. $\bigcap[-\frac{1}{n}, \frac{1}{n}] = \{0\}$: 0 is in every interval. If $x \neq 0$, choose $n > \frac{1}{|x|}$; then $|x| > \frac{1}{n}$, so $x \notin [-\frac{1}{n}, \frac{1}{n}]$. (Or apply NIT(b): the lengths $\frac{2}{n} \rightarrow 0$.)

$\bigcap(0, \frac{1}{n}] = \emptyset$: if $x \leq 0$ it is in none of them. If $x > 0$, choose $n > \frac{1}{x}$, so $x > \frac{1}{n}$ and $x \notin (0, \frac{1}{n}]$. This doesn't contradict NIT because the intervals are *not closed*: the point they shrink to, 0, has been removed. $\square$

36. *Nested*: $1 - \frac{1}{n} \leq 1 - \frac{1}{n+1}$ and $1 + \frac{1}{(n+1)^2} \leq 1 + \frac{1}{n^2}$, so $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$, which gives $I_{n+1} \subseteq I_n$. *Intersection*: $1 \in I_n$ for all n . The lengths $\frac{1}{n} + \frac{1}{n^2} \leq \frac{2}{n} \rightarrow 0$, so by NIT(b) the intersection is exactly $\{1\}$. $\square$

37. Both x_n and α are in $[a_n, b_n]$, so $|x_n - \alpha| \leq b_n - a_n$. Given $\varepsilon > 0$, choose N with $b_n - a_n < \varepsilon$ for $n \geq N$. Then $|x_n - \alpha| < \varepsilon$ for $n \geq N$. $\square$

12 Practice Exam 1 (75 minutes)

Set a 75-minute timer, use no notes, and write full proofs. Solutions are in the next section.

Exam Problem 1 (Definitions, 10 pts). State precisely: (a) $\sup S$; (b) $x_n \rightarrow L$; (c) (x_n) is Cauchy, and its negation; (d) the Completeness Axiom; (e) the Nested Interval Theorem.

Exam Problem 2 (Induction, 15 pts). Prove that $\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}$ for all $n \in \mathbb{N}$.

Exam Problem 3 (Sup/inf, 15 pts). Let $A \subseteq (0, \infty)$ be nonempty with $a = \inf A > 0$. Let $B = \{\frac{1}{x} : x \in A\}$. Prove $\sup B = \frac{1}{a}$.

Exam Problem 4 (ϵ - N , 15 pts). Prove from the definition that $\lim_{n \rightarrow \infty} \frac{n^2 + 2n}{3n^2 - 1} = \frac{1}{3}$.

Exam Problem 5 (Monotone sequences, 15 pts). Let $x_1 = 1$ and $x_{n+1} = \sqrt{3 + 2x_n}$. Prove (x_n) converges and find its limit.

Exam Problem 6 (Cauchy sequences, 20 pts).

- (a) Prove that every convergent sequence is Cauchy.
- (b) Let $s_n = \sum_{k=1}^n \frac{\cos k}{k^2}$. Prove (s_n) converges. (Note: (s_n) is not monotone.)
- (c) Let $u_n = \sum_{k=1}^n \frac{1}{2k-1}$. Prove (u_n) is not Cauchy.

Exam Problem 7 (True/False, 10 pts). Justify each answer briefly or give a counterexample.

- (a) If $x_n < 1$ for all n and $x_n \rightarrow L$, then $L < 1$.
- (b) If (x_n) is increasing and not bounded above, then $x_n \rightarrow \infty$.
- (c) If $(|x_n|)$ converges, then (x_n) converges.
- (d) $\sup\{r \in \mathbb{Q} : r < \sqrt{2}\} = \sqrt{2}$.
- (e) If (x_n) converges and (y_n) diverges, then $(x_n + y_n)$ diverges.
- (f) If I_n are nonempty, nested and bounded intervals, then $\bigcap_n I_n \neq \emptyset$.

13 Practice Exam 1: solutions

1. (a)–(d): see the definitions cheat sheet. Negation of Cauchy: $\exists \varepsilon > 0 \forall N \exists n, m \geq N$ with $|x_n - x_m| \geq \varepsilon$. (e): see Theorem 8.20. Say “closed, bounded, nested” explicitly.

2. $P(n)$: $\sum_{k=1}^n \frac{1}{k^2} \leq 2 - \frac{1}{n}$. Base: $1 \leq 2 - 1$. Assume $P(n)$. Then $\sum_{k=1}^{n+1} \frac{1}{k^2} \leq 2 - \frac{1}{n} + \frac{1}{(n+1)^2}$, and we need this to be $\leq 2 - \frac{1}{n+1}$, i.e. $\frac{1}{(n+1)^2} \leq \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$. This holds because $n(n+1) \leq (n+1)^2$. $\square$

Lesson: trying to prove “ ≤ 2 ” directly by induction fails, because the hypothesis is too weak. The stronger statement is easier to prove.

3. *Upper bound:* for $x \in A$, $x \geq a > 0$, so $\frac{1}{x} \leq \frac{1}{a}$. So $\frac{1}{a}$ is an upper bound of B and $\sup B \leq \frac{1}{a}$. *Least* (Quiz 1 style, no ε): let u be any upper bound of B . Then $u > 0$, since B has positive elements. For each $x \in A$, $\frac{1}{x} \leq u$, so $x \geq \frac{1}{u}$. So $\frac{1}{u}$ is a lower bound of A , and $\frac{1}{u} \leq a$ (a is the *greatest* lower bound). Hence $u \geq \frac{1}{a}$. So $\frac{1}{a}$ is the least upper bound. $\square$ (If $\inf A = 0$, then B is unbounded.)

4. $\frac{n^2+2n}{3n^2-1} - \frac{1}{3} = \frac{6n+1}{3(3n^2-1)} > 0$. For $n \geq 1$: $6n+1 \leq 7n$ and $3n^2-1 \geq 2n^2$, so this is $\leq \frac{7n}{6n^2} = \frac{7}{6n}$. Let $\varepsilon > 0$ and choose $N > \frac{7}{6\varepsilon}$. For $n \geq N$: $|\frac{n^2+2n}{3n^2-1} - \frac{1}{3}| \leq \frac{7}{6n} \leq \frac{7}{6N} < \varepsilon$. $\square$

5. *Bounds:* $0 < x_n < 3$ by induction. True for $n = 1$, and $0 < x_n < 3$ gives $3 < 3 + 2x_n < 9$, so $0 < \sqrt{3} < x_{n+1} < 3$. *Increasing:* $x_{n+1}^2 - x_n^2 = 3 + 2x_n - x_n^2 = (3 - x_n)(1 + x_n) > 0$, and both terms are positive, so $x_{n+1} > x_n$. By MCT, $x_n \rightarrow L$, and $L \geq x_1 = 1$. Square the recurrence: $x_{n+1}^2 = 3 + 2x_n$. By limit laws, $L^2 = 3 + 2L$, so $(L - 3)(L + 1) = 0$. Since $L \geq 1$, $L = 3$. $\square$

6. (a) Part I of Theorem 8.14 (the $\frac{\varepsilon}{2}$ argument). (b) For $m > n$: $|s_m - s_n| = |\sum_{k=n+1}^m \frac{\cos k}{k^2}| \leq \sum_{k=n+1}^m \frac{1}{k^2} < \frac{1}{n}$, by Example 8.7. Choose $N > \frac{1}{\varepsilon}$. So (s_n) is Cauchy, and by the Cauchy criterion it converges. (MCT doesn’t apply, since the terms $\cos k$ change sign.) (c) Take $\varepsilon = \frac{1}{4}$. Given N , let $n = N$, $m = 2N$. The sum $u_{2N} - u_N = \sum_{k=N+1}^{2N} \frac{1}{2k-1}$ has N terms, each $\geq \frac{1}{4N-1} > \frac{1}{4N}$, so $u_{2N} - u_N > \frac{1}{4}$. $\square$

7.

- (a) False: $1 - \frac{1}{n} \rightarrow 1$.
- (b) True: given M , M is not an upper bound, so some $x_N > M$. For $n \geq N$, $x_n \geq x_N > M$.
- (c) False: $(-1)^n$.
- (d) True: $\sqrt{2}$ is an upper bound. Given $\varepsilon > 0$, density gives $r \in \mathbb{Q}$ with $\sqrt{2} - \varepsilon < r < \sqrt{2}$.
- (e) True: if $x_n + y_n$ converged, then $y_n = (x_n + y_n) - x_n$ would converge.
- (f) False: $(0, \frac{1}{n})$ are not closed and have empty intersection. (True if the intervals are closed.)